Marginal Utility Shocks and the Precautionary Saving Puzzle

Mohammad Amin Shirazian[1]
Toronto Metropolitan University

Latest version

Abstract

Documenting that both saving and medical spending accelerate at the top, the paper derives a sufficient condition for saving to remain positive at arbitrarily high wealth and shows that the resulting stationary wealth distribution has Pareto tails, the first such result from precautionary saving alone. Calibrated to the data, the mechanism accounts for roughly 11 percent of observed saving in the top wealth decile.

Empirically, it documants that saving among pre-retirement couple households in the HRS is positive throughout the wealth distribution and rises sharply at the top, reaching more than $1 million in the top wealth decile. Medical spending follows a similar pattern: households in the top wealth decile spend, on average, 82.5 percent more on medical care than households in the bottom decile. Calibrated to these facts, the model predicts saving of $121,960 against $1,085,280 observed in the top decile—a ratio of about 0.11, so that the precautionary health-shock channel alone accounts for roughly 11 percent of observed average saving among the wealthiest couples. By contrast, a standard incomplete-markets model without this channel predicts dissaving of −$1,992,990 at the same decile, a gap of approximately $2.1 million.

Theoretically, the paper establishes two results. First, the paper derives a sufficient condition for the precautionary saving motive to remain active at arbitrarily high wealth: as long as the marginal utility of consumption drops faster than the marginal return on self-insuring against mortality shocks, saving does not turn negative beyond some wealth threshold. Second, it proves that the wealth distribution implied by the mechanism has Pareto tails. Given that saving remains positive beyond this threshold and mortality generates random separation across cohorts, the resulting stationary wealth distribution follows Zipf's law—a Pareto tail with unit exponent—at the top.

---

[1]Toronto Metropolitan University. I thank my advisor, Keyvan Eslami, and Claustre Bajona and Kevin Fawcett, for their guidance and support.

1 Introduction

Does health spending play a role in saving behavior at the top of the wealth distribution? A tension exists between the literature that characterizes health as a luxury good (Hall and Jones, 2007) and models that explain wealth heterogeneity through medical spending risk (DeNardi and Fella, 2017). The latter class of models assume that the incentive to allocate resources toward health-related consumption weakens as wealth rises[2]. By contrast, the former suggests that, because survival remains valuable even at higher levels of wealth, households continue to increase health-related expenditures as wealth grows, thereby generating luxury-good behavior in health.

Theoretically, this paper bridges this gap in the literature through a twofold contribution. First, it derives sufficient conditions under which the precautionary saving motive remains active at the top of the wealth distribution when a luxury-health framework is incorporated into an income-fluctuation model.[3] Second, it proves the existence of a stationary wealth distribution with Pareto tails. These findings contribute to the class of models that seek to explain wealth concentration at the top of the wealth distribution through non-precautionary channels—bequests, human capital transfers, preference heterogeneity, richer earnings processes, heterogeneous returns, and entrepreneurship—being the first to argue that precautionary saving contributes to wealth concentration at the top of the wealth distribution.[4]

The paper has several empirical contributions. First, it documents that, contrary to the prediction of Bewley models (Bewley, 1977), saving rises with wealth among pre-retirement couples in the HRS. Among couples, saving is positive and increasing throughout the distribution, ranging from \$23,000 to \$45,000 in the bottom seven deciles, rising to \$71,000 in the eighth decile and \$78,000 in the ninth, and reaching \$1,092,000 in the top decile. It further confirms that the survey saving data are in line with administrative data, and documents the precautionary saving puzzle in the HRS.

Second, it documents that the key condition under which health becomes a luxury good is satisfied for pre-retirement households in the Health and Retirement Study. The condition depends on

[2]In this class of models, health spending follows an exogenous stochastic process. Consequently, as net worth grows, households become increasingly able to self-insure against realizations of medical spending risk.

[3]The luxury-health framework we use is Hall and Jones (2007), though other frameworks could serve the same purpose.

[4]Refer to the Literature Review section for an extended discussion.

the intertemporal elasticity of substitution and the elasticity of health with respect to medical spending.[5] The latter parameter is estimated by matching model-implied survival probabilities to self-reported survival expectations and medical spending in the Health and Retirement Study using nonlinear least squares, while the former is calibrated from Hosseini (2015). The estimated parameters indicate that the marginal gain from allocating resources to self-insurance against mortality shocks remains consistently higher than the marginal utility increments from additional consumption throughout the wealth distribution, thereby making health a luxury good.

Finally, the study reconciles model-implied saving with observed saving. Against observed average saving of roughly $1.1 million, the model predicts saving of $121,960, whereas the standard income fluctuation model predicts dissaving of $1,992,990. Thus, the model closes nearly $2.1 million of the gap between observed saving and the benchmark prediction. This degree of explanatory power, derived solely from precautionary saving in response to marginal utility shocks, is quantitatively significant. It suggests that risk to future utility, particularly from health-related uncertainty, may be an important driver of upper-tail saving, complementing alternative explanations.

In what follows, Section 1.1 reviews the literature on wealth concentration and the rising share of health expenditure in GDP. Section 2 presents the motivation for the analysis. Section 3 introduces the theoretical framework, while Section 4 establishes the conditions under which a stationary distribution exists. Section 5 estimates the survival function and calibrates the model to compare predicted and observed savings. Section 7 concludes.

1.1 Related Literature

Standard macroeconomic models with complete markets, such as those in Ljungqvist and Sargent (2012), do not generate precautionary saving. With complete markets, households can perfectly smooth consumption against future uncertainty, eliminating the need to self-insure. Moreover, when agents are homogeneous in initial conditions, these models provide no internal mechanism through which wealth heterogeneity can emerge.

A major departure from this benchmark is the class of incomplete-markets models introduced by

[5] Explicitly, the condition is $1-\sigma < -\beta$, where $1-\sigma$ governs the curvature of utility and $-\beta$ determines the sensitivity of mortality risk to health.

Bewley (1977) and further developed by Aiyagari (1994) and Huggett (1993), commonly referred to as BAH models. These models introduce market incompleteness and idiosyncratic income risk, so that an individual's wealth depends on the entire history of realized shocks. The absence of full insurance creates a precautionary saving motive against uninsurable income risk, generating wealth dispersion. As a result, BAH models have become workhorse models of inequality in modern macroeconomics, closely related to the income fluctuation models of Schechtman (1976); Schechtman and Escudero (1977).

However, BAH models have difficulty accounting for inequality at the top of the wealth distribution. Empirically, wealth distributions exhibit Pareto tails in the United States (Saez and Zucman, 2016), Canada (Statistics Canada, 2025), and Europe (OECD, 2024). By contrast, even when calibrated with highly skewed idiosyncratic income risk, standard BAH models tend to generate thin-tailed wealth distributions. The reason is that, even in the absence of Arrow securities, sufficiently wealthy households can smooth consumption almost perfectly through asset accumulation. Once wealth is high enough, the precautionary saving motive fades, and saving eventually becomes negative. Life-cycle versions of BAH models reach a similar conclusion (Huggett, 1996).

To address this limitation, subsequent extensions introduce additional channels capable of generating upper-tail wealth concentration (De Nardi and Fella, 2017). These include luxury bequests (De Nardi, 2004), heterogeneous time preferences (Cagetti, 2003), richer earnings risk (Castaneda et al., 2003), medical expense risk (De Nardi et al., 2010), heterogeneity in rates of return (Benhabib et al., 2011, 2015), and entrepreneurship (Cagetti and De Nardi, 2006). These mechanisms introduce new saving motives at the top of the wealth distribution. Finally, De Nardi and Fella (2017) emphasizes the importance of endogenizing additional decision variables that may themselves shape saving behavior. This point is especially relevant for models that seek to incorporate the luxury nature of health, since health spending must respond endogenously to household resources for the luxury channel to operate.

In this class of wealth-distribution models, health spending is typically treated as an exogenous cost, with the motive to save for it vanishing as wealth rises. The literature on the rising share of health spending in GDP, however, tells a different story. Emphasizing a survival mechanism, Hall and Jones (2007) argue that health can become a luxury good because health affects the value of consumption itself. A unit of consumption is not equally valuable across health states: the marginal value of consumption depends on whether the household is healthy or sick. The relevant decision is therefore not only how to smooth consumption across states, but also how to preserve the health state in which consumption remains valuable. Hall and Jones (2007) use this mechanism to account for the rising share of health spending in U.S. GDP.

Empirically, a large literature documents that health spending rises with income and wealth, both across households and in the aggregate. Using MEPS data, Ales et al. (2014) show that health care increases survival probabilities, while Ozkan (2014) emphasizes the distinction between health investment among the rich and medical spending among the poor. French and Kelly (2016) document the cross-country concentration of high medical spending in high-utilization states. Dickman et al. (2016, 2017) further document a widening gap in medical spending across income groups.

2 Motivation

2.1 Saving Acceleration

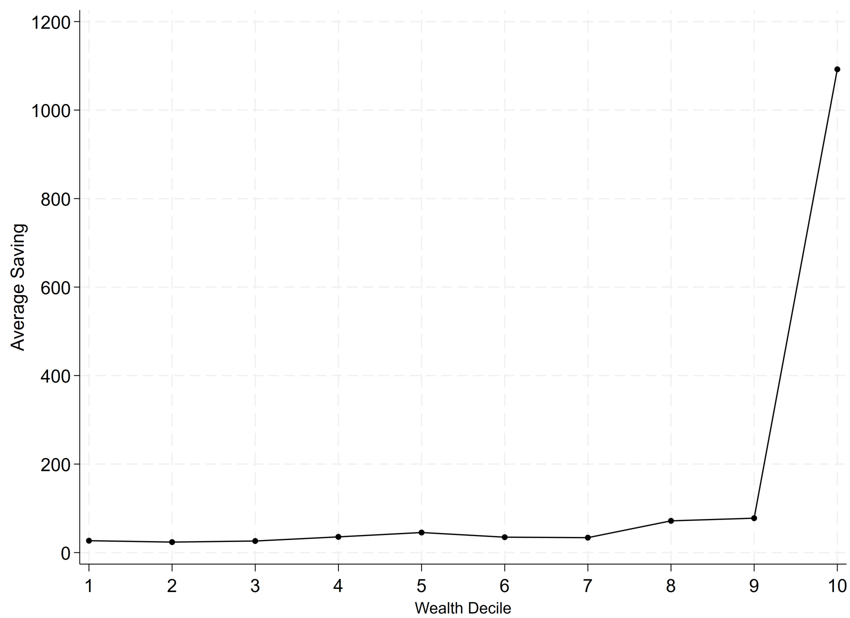
As documented by the data appendix, contrary to the prediction of Bewley models (Bewley, 1977), saving rises with wealth among pre-retirement couples in the HRS. Figure 1 reports average savings of couple households in different wealth brackets among 55-to-64-year-old U.S. households from 1996 to 1998 in thousands of dollars where saving is defined by the first difference of wealth. The first seven wealth groups report savings between $23,700 and $45,400, with these figures rising to $71,800 and $77,700 for the eighth and ninth wealth groups, respectively. A substantial increase in savings is observed for the wealthiest couples, with households in the top group accumulating approximately $1,092,000 in net worth on average. [6]

The main question here is what drives this acceleration of saving at the top. As discussed previously by (Fagereng et al., 2019; Bach et al., 2017) using Swedish and Norwegian administrative data, higher returns associated with a higher stock of wealth are the main underlying factor. Table A3, panel b, also observes a similar pattern. Couple households at the top of the wealth distribution

[6] Appendix A explains the cohort selection process, wealth distribution, and saving patterns of singles in addition to couples.

have a more diversified portfolio including financial, housing, pension, transport, and business wealth. Middle households, however, mostly depend on their primary housing wealth, which is almost 50% of their wealth. Additionally, Table A6 reports that the correlation of saving rates and earnings is weak across wealth deciles while the correlation of saving rates and capital income rises in the higher deciles of wealth.

FIGURE 1. Average Saving



Notes: This figure represents average savings of couple households in different wealth brackets among the US elderly from 1996 to 1998. Average savings are in thousands of dollars.

Rather than saving, one could focus on the wealth distribution implied by Bewley models. As surveyed by De Nardi and Fella (2017), wealth is highly concentrated at the top and follows a Pareto distribution: the top 1% of households hold more than 30% of wealth, while the top 5% hold more than 50% of total assets. Consequently, theoretical contributions propose mechanisms that generate a stationary distribution with Pareto tails and high concentration of wealth at the top.

Whether framed as high saving or high concentration, both narratives assume that the precautionary motive cannot explain continuous accumulation of wealth and its increasing concentration at the top. Therefore, alternative channels have been introduced to address this limitation. The literature on the rising share of health spending in GDP, however, tells a different story. Hall and Jones

(2007) suggests that, because survival remains valuable even at higher levels of wealth, households continue to endogenously increase health-related expenditures as wealth grows, implying the persistence of the precautionary motive at the top.

The central difference between these two branches lies in how they incorporate health shocks. In the former class of models, health spending follows an exogenous stochastic process. Consequently, as net worth grows, households become increasingly able to self-insure against realizations of medical spending risk. By contrast, the latter suggests that, because survival remains valuable even at higher levels of wealth, households continue to increase health-related expenditures as wealth grows, thereby generating luxury-good behavior in health.

2.2 Medical Spending Acceleration

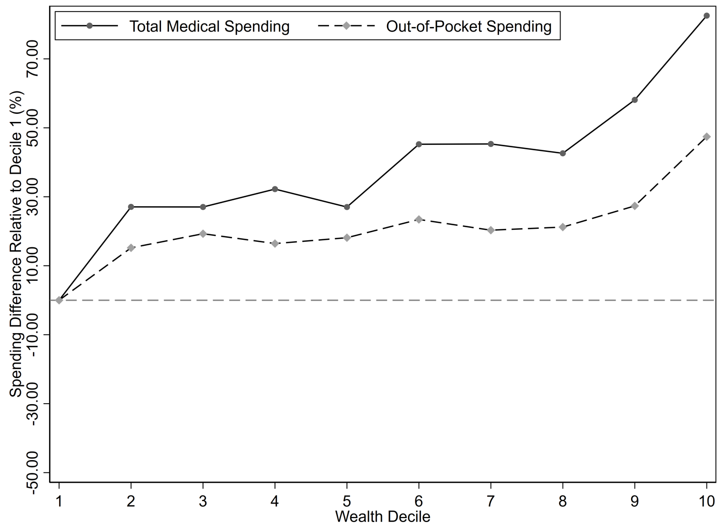
Like saving, medical spending accelerates for couple households at the top of the wealth distribution. This is the second key empirical fact motivating our theoretical framework.

Figure 2 shows that the total imputed medical-spending gradient rises from 27 percent at the second decile to 82 percent at the tenth decile relative to the poorest households, with a sharp 24-percentage-point jump between deciles 9 and 10. The out-of-pocket gradient ranges from 15 to 27 percent across deciles 2 through 9 and rises to 47 percent at the top decile—also a sharp jump, of 20 percentage points, between deciles 9 and 10. Both measures therefore confirm the same sharp acceleration in the top decile.[7]

This raises the paper's central questions. First, can we prove that incorporating Hall and Jones (2007)'s framework into an income-fluctuation setup guarantees saving remains positive beyond some wealth threshold? Second, can we prove that it generates a stationary wealth distribution with Pareto tails? Third, quantitatively, how much of the observed saving of top-decile households can this mechanism explain?

[7] Appendix B reports the imputation method and the analogous results for single-person households, along with the full imputation methodology.

FIGURE 2. Total vs. Out-of-Pocket Medical Spending by Wealth Decile—Couples



Notes: Estimated percentage difference in medical spending relative to decile 1 for couple households, using imputed total medical expenditure (solid) and observed out-of-pocket expenditure (dashed). See Appendix B for details.

3 Saving at the Tail

To understand the reason our framework can give rise to a saving function that is increasing in wealth, after some threshold—in contrast to the BAH's framework where saving becomes negative after some threshold—we consider two stylized environments with uninsurable idiosyncratic shocks in this section: An income fluctuations problem, where individuals have to self-insure against fluctuations in their income, and a modified framework where income shocks are replaced by mortality shocks—or, more accurately, shocks to continuation marginal utility.

Consider the problem of an individual in a standard income fluctuations framework, where income can take one of two values, $\{\underline{y}, \bar{y}\}$, where $\underline{y} < \bar{y}$. If we refer to these two as the "low" and "high states," and denoting the value and policy functions in each state by means of lower and upper bars, respectively, we can write the individual's problem in high state as

$$\bar{V}(b) = \max_{c,b'} \quad \left\{ u(c) + \rho \left[\pi \bar{V}(b') + (1-\pi) \underline{V}(b') \right] \right\} \tag{BAH}$$
$$s.t. \quad b' = (1+r)\, b + \bar{y} - c,$$
$$b' \geq \underline{b}.$$

Here, c represents total consumption, and the utility function, $u(c)$, typically takes a CRRA form. Total future wealth, b', is the residual amount of total resources not consumed today. In this setting, b' and c are the choice variables, while b and y are the state variables, representing current total assets and income. Because the framework is partial equilibrium, the interest rate r is determined exogenously. The parameter ρ is the household's discount factor, governing how it values future utility relative to present utility. In this problem, π is the conditional probability of staying in the high state in the next period, conditioned on being in the high state in the current period. [8]

In our framework, we replace income shocks by mortality shocks. To this end, we denote an individual's underlying health by h, and assume it can take two values, , $\left\{ \underline{h}, \bar{h} \right\}$, where $\underline{h} < \bar{h}$. With some abuse of terminology, we will refer to these two as the "low" and "high states," and keep denoting them via lower and upper bars as in problem (BAH).[9]

To make sure the proceeding arguments deal explicitly and entirely with these two different mechanisms, in this section, we assume that the low state is an absorbing state in this latter framework: Once an individual enters the low state, he remains there forever. The probability of an individual transitioning to this state from the high state is denoted by $1-\pi$, as in problem (BAH). In addition, we assume that an individual does not face any chance of mortality, whereas this

[8]The problem in the low state is rather similar to (BAH):

$$\underline{V}(b) = \max_{c,b'} \quad \left\{ u(c) + \rho \left[\pi_l \bar{V}(b') + (1-\pi_l) \underline{V}(b') \right] \right\}$$
$$s.t. \quad b' = (1+r)\, b + \underline{y} - c,$$
$$b' \geq \underline{b}.$$

Here π_l is the probability of moving to the high state while the low state has realized today.

[9]This choice of notation serves a simplifying purpose, as it becomes apparent when writing the first order conditions. It should be clear which problem a function refers to from the context. We make the reference explicit if not.

probability in the low state is determined by the *health production function*, $\chi(m)$, where m captures what the individual spends on his health.

As such, the individual's problem in the high state is quite similar to that in (BAH) in this alternative setting:[10]

$$\begin{aligned}\bar{V}(b) = \max_{c,b'} \quad & \left\{u(c) + \rho\left[\pi\bar{V}(b') + (1-\pi)\underline{V}(b')\right]\right\} \\ s.t. \quad & b' = (1+r)b + y - c, \\ & b' \geq \underline{b},\end{aligned} \tag{BL-H}$$

where y denotes the individuals' income, which is independent of their health state and deterministic(so that we can focus on marginal utility shocks, as stated earlier). In the low state, this problem becomes:

$$\begin{aligned}\underline{V}(b) = \max_{c,m,b'} \quad & \left\{u(c) + \rho\cdot\chi(m)\cdot\underline{V}(b')\right\} \\ s.t. \quad & b' = (1+r)b + y - c - m, \\ & b' \geq \underline{b}.\end{aligned} \tag{BL-L}$$

Note that, in this problem, we have implicitly normalized the value upon death to zero.[11]

In all that follows, we are going to assume that the Bernoulli utility takes the CRRA functional form with a constant term representing the value of being alive:

$$u(c) = \nu + \frac{c^{1-\sigma}}{(1-\sigma)},$$

for some $\sigma > 0$ and $\nu > 0$ such that, for some minimal sustainable level of consumption, $\underline{c}$, we have $u(\underline{c}) > 0$.[12] Moreover, we assume the following form for the health production function for

---

[10]The consumer's health state determines which problem is solved. A healthy consumer faces problem BL-H, while a consumer in the high-utilization state solves problem BL-L.

[11]With probability $1-\chi(m)$ the consumer faces mortality and receives $\underline{V}(b') = 0$

[12]Otherwise, (BL-L) turns into a trivial problem where individuals prefer death to being alive. The CRRA utility form is indispensable to Huggett (1993)'s proof of negative saving above some level.

the sake of tractability:

$$\chi(m) = 1 - \left(\frac{1}{\alpha \cdot m + \underline{h}}\right)^{\beta}, \tag{1}$$

for some $\alpha > 0$ and $\beta \leq 1$. for some $\alpha > 0$ and $\beta \leq 1$. for some $\alpha > 0$ and $\beta \leq 1$.[13]

Consider the Euler equation for an *interior solution* to problems (BAH) and (BL-H):[14]

$$\bar{V}_b(b) = \rho \cdot (1+r) \cdot \left[\pi \bar{V}_b(b') + (1-\pi)\underline{V}_b(b')\right], \tag{HEE}$$

where subscripts show first order derivatives, and we have used the envelope condition to replace for marginal utility of consumption:

$$\bar{V}_b(b) = (1+r) \cdot u_c(c). \tag{HEC}$$

Under the assumption that the value function is concave in wealth in both states, the Euler equation determines the direction of saving at any current level of wealth. To see this more clearly, consider momentarily the case in which $\pi = 1$. When $\rho(1+r) = 1$, we have $b' = b$, implying zero saving and perfectly smooth consumption: $c = c' = rb + \bar{y}$ (or $c = c' = rb + y$ when income is constant). In other words, when the rate of time preference and the rate of return exactly offset one another, the individual perfectly smooths consumption over time.

---

[13] Intuitively, equation 1 can be written as $\chi(m) = 1 - [h(m;\alpha)]^{-\beta}$. Here, αm can be interpreted as effective medical spending, with α capturing the efficiency with which medical spending is transformed into health. The parameter β governs the sensitivity of survival to improvements in health.

[14] The derivation of the Euler condition is straightforward. For an interior solution, the first-order condition is

$$u_c(c(b)) = \rho\left[\pi \bar{V}_b(b'(b)) + (1-\pi)\underline{V}_b(b'(b))\right].$$

And the budget constraint implies:

$$c'(b) + b'(b) = (1+r)$$

Moreover, equation BL-H implies

$$\bar{V}_b(b) = \left\{u_c(c(b)) + \rho\left[\pi \bar{V}_b(b'(b)) + (1-\pi)\underline{V}_b(b'(b))\right]\right\}(c'(b) + b'(b)).$$

Implicit differentiation of the budget constraint, together with substitution of the first-order condition for u_c, yields the result.

Under the standard assumption that the interest rate is not sufficiently high to offset time preference (Aiyagari, 1994), so that $\rho(1+r) < 1$, the individual front-loads consumption. In this case, for equation (HEE) to hold, $\bar{V}_b(b')$ must exceed $\bar{V}_b(b)$. Concavity of the value function then implies $b' < b$. Consequently, in the lack of income shocks, saving should be negative. [15]

In the presence of uninsurable risk, saving need not be negative at lower levels of wealth even when $\rho\left(1+r\right) < 1$, provided that the realization of a shock—whether to income or underlying health—substantially increases the marginal value of assets in the future. We refer to this term, $V_b\left(b'\right)$, as the *marginal continuation value.*

$$\pi\left[\frac{\bar{V}_b\left(b'\right)}{\bar{V}_b\left(b\right)}\right] + (1-\pi)\left[\frac{\underline{V}_b\left(b'\right)}{\bar{V}_b\left(b\right)}\right] = \frac{1}{\rho\cdot(1+r)}. \tag{2}$$

When the value function is concave, the first term in brackets in this equation will be less than one when $b' > b$. Therefore, the only possibility for this equality to hold when $b' > b$ and $\rho\left(1+r\right) < 1$ is for $\underline{V}_b\left(b'\right)$ to be greater than $\bar{V}_b\left(b\right)$ by a "large enough" margin, and this is the logic behind the terminology "precautionary saving": If the value of having one more unit of wealth is high enough *when an unfavourable shock is realized*, individual tends to save.

One can summarize this argument as follows: Under the concavity assumption for the value function, saving is positive at any current level of wealth b if, and only if,

$$\pi\left[\frac{\bar{V}_b\left(b\right)}{\bar{V}_b\left(b\right)}\right] + (1-\pi)\left[\frac{\underline{V}_b\left(b\right)}{\bar{V}_b\left(b\right)}\right] > \frac{1}{\rho\cdot(1+r)}.$$

Similarly, saving will be zero or negative, respectively, when wealth is equal to b if, and only if, this inequality holds as strict equality or the left-hand side is strictly less than the constant term on the right-hand side. This inequality can be rearranged as

$$(1-\pi)\left[\frac{\underline{V}_b\left(b\right) - \bar{V}_b\left(b\right)}{\bar{V}_b\left(b\right)}\right] > \frac{1}{\rho\cdot(1+r)} - 1. \tag{3}$$

[15] Trivially, when $\rho(1+r) \geq 1$, the model implies unbounded asset accumulation (Aiyagari, 1994; Schechtman, 1976). However, this case is generally viewed as unrealistic, as it requires the interest rate to be sufficiently high that individuals prefer future consumption to current consumption.

The right-hand side of this equality is a positive constant when $\rho(1+r) < 1$[16]. As such, the shape of optimal saving as a function of current wealth, b, is determined by the marginal value of wealth in the low state *relative to that in the high state* (and not by the difference between the two).

This is rather intuitive: An individual saves as a "precaution" against the possibility of low shocks, and as long as the marginal benefit of one additional unit of saving is high enough should the low shock arrive. Saving is a precaution against fluctuations in the marginal continuation value when the value function is concave. On the other hand, when the ratio of marginal continuation value in the low state to that in the high state tends to one, this precautionary saving motive vanishes: When there is not much fluctuation in the marginal continuation value in the first place, there are no incentives to insure against such fluctuations either.[17]

One can use condition (HEC) to write (3) as

$$(1-\pi)\left[\frac{u_c(\underline{c}(b)) - u_c(\bar{c}(b))}{u_c(\bar{c}(b))}\right] > \frac{1}{\rho\cdot(1+r)} - 1, \tag{4}$$

where $\underline{c}(\cdot)$ and $\bar{c}(\cdot)$ are the policy functions in low and high states, respectively. If we replace the utility function with its CRRA form, we can write this inequality as:

$$\left[\frac{\bar{c}(b)}{\underline{c}(b)}\right]^{\sigma} > \left[\frac{\rho^{-1}(1+r)^{-1} - \pi}{1-\pi}\right]. \tag{5}$$

Note that the first term of the numerator on the right-hand side of this inequality is greater than one. As such, the right-hand side is a constant, greater than one.

To see the intuition behind this inequality, we can use a linear expansion around $\bar{c}(b)$ to write the numerator in (4) in terms of consumption in two states:

$$(1-\pi)\left[\frac{u_{cc}(\bar{c}(b))\left[\underline{c}(b) - \bar{c}(b)\right]}{u_c(\bar{c}(b))}\right] > \frac{1}{\rho\cdot(1+r)} - 1.$$

---

[16] As discussed previously, this condition holds under the standard assumptions.

[17] This is not to say that the only incentive for saving is precautionary motive. The ratio of time discount rate to the risk-free rate is still a strong determinant of saving behaviour. Moreover, the precautionary saving motive is determined by the probability of the realization of unfavourable shocks.

Using the CRRA utility form, the ratio on the left-hand side of this inequality can be written as:

$$1 - \frac{\underline{c}\,(b)}{\bar{c}\,(b)} > \frac{1 - \rho \cdot (1 + r)}{\sigma \cdot (1 - \pi) \cdot \rho \cdot (1 + r)}.^{18} \tag{6}$$

As such, the shape of saving function depends critically on σ, π, and $\rho\,(1 + r)$, as well as the ratio $\underline{c}\,(b)\,/\bar{c}\,(b)$ given these parameters. As this inequality suggests, if $1 - \pi$ or σ are too small, it is well possible for saving to remain negative at all levels of wealth. This is rather intuitive: When the possibility of a negative shock is "too small," or when individuals are "too risk neutral," precautionary saving motive would also be too small. In such cases, if the incentive to front-load consumption is strong enough—when $\rho\,(1 + r)$ is far from unity—individual might not have any incentives to save against unlikely adverse shocks in the future.

On the other hand, when $1 - \pi$ and σ are large enough, it is possible for the precautionary saving motive to dominate the motive to front-load consumption, leading to positive saving. When this is the case, we'd expect the ratio of consumption in two states to remain far from one. The same forces that determine how this ratio changes as wealth increases (in the optimal solution) also determine the shape of optimal saving as a function of wealth.

The aforementioned arguments are summarize in Lemma 3.1, under the assumption that the solutions to the functional equations in (BL-H) and (BL-L) are concave. This assumption, however, cannot be proved analytically: The product of two concave functions on the right-hand side of functional equation (BL-L) is not necessarily concave. As a result, the usual inheritance arguments in dynamic programming do not carry over to this case. As such, throughout this section, we maintain the following assumption:

Assumption 3.1 *The solution to the functional equation* (BL-L), $\underline{V}\,(\cdot)$*, is concave.*

Note that, if $\underline{V}\,(\cdot)$ is concave, $\bar{V}\,(\cdot)$ will necessarily be concave. Moreover, we know from the envelope condition that assumption 3.1 is equivalent to the policy function, $\underline{c}\,(\cdot)$, being increasing.[19] In our numerical experiments, we have not managed to come up with a set of parameter values

[18]Note that, since $u_c\,(\cdot)$ itself is a convex function, this inequality is the sufficient condition for the saving rate to be positive and not a necessary condition, at least unless consumption is not "too large."

[19]From the Eulor Condition $V_b(b) = (1 + r)\,u_c(c(b))$ is decreasing in c, if $c(.)$ is increasing in b then $V(.)$ is concave. The only if part is trivial to show.

that violate this latter condition.

As a simple example, assume that the consumer does not save—$b'(b) = b$—and $\rho = 1$ the dynamic programing problem simplifies to:

$$\underline{V}(b) = \max_{c,m} u(c) + \chi(m)\underline{V}(b) \qquad \text{s.t.} \qquad c + m = y + rb$$

To see this for a simple static model, where $h(m) = (\underline{h} + am)^{\beta}$, one can simplify our problem to:

$$V(b) = \max_{c,m} h(m)\, u(c) \qquad \text{s.t.} \qquad c + m = y + rb$$

Since $h(.)$ and $u(.)$ are both concave functions, the joint concavity of $h(.)\, u(.)$ shrinks to:

$$\left(-\frac{h(m)h''(m)}{[h'(m)]^2}\right)\left(-\frac{u(c)u''(c)}{[u'(c)]^2}\right) > 1.$$

Applying the functional forms, the condition transtlates to:

$$\frac{1-\beta}{\beta}\,\sigma\left(vc^{\sigma-1} + \frac{1}{1-\sigma}\right) > 1$$

Because endogenous medical spending multiplies the continuation value through the survival function, standard concavity-preservation results do not apply directly: the product of two positive concave functions need not be jointly concave. We therefore maintain concavity of the low-health value function as an explicit assumption. This property is satisfied by the calibrated numerical solution, as verified by the monotonicity of the consumption policy and the nonincreasing derivative of the value function over the asset grid.

Under the calibrated values, discussed int the section 5[20], this condition reduces to $c > 0.067$. Hence, the objective is jointly concave over the economically relevant consumption range.

To see the application in the infinite period problem, consider the discrete form:

[20] $\beta = 0.4$, $\sigma = 2$, and $v = 20$

Lemma 3.1 *Under assumption 3.1 and the maintained differentiability and utility assumptions, optimal saving in the high state of* (BAH) *or* (BL-H) *is positive at wealth* b *if the marginal value functions satisfy* (3).

Proof. If condition (3) held with equality, this choice would satisfy the optimal saving condition. Since the left-hand side exceeds the right-hand side, $\underline{V}_b(b' = b)$ must decline to restore equality, since all other terms are parameters or current-state marginal value functions evaluated at the given current state. By assumption 3.1, $\underline{V}$ is concave, so $\underline{V}_b$ is nonincreasing. A decline in this marginal value therefore requires an increase in b' above b, implying positive saving. Thus, condition (3) is sufficient for positive saving. □

The preceding argument suggests that, to determine the shape of saving as a function of current wealth in either problem, one can focus on the ratio $\bar{c}\,(b)\,/\underline{c}\,(b)$ and how it changes with wealth. One can show that, in problem (BAH), this ratio converges to one as $b \to \infty$, and, as such, there exists some $\bar{b}$ above which saving inevitably becomes negative.[21]

In our environment with health shocks, on the other hand, the same mechanics but in the opposite direction indicate the exact inverse regarding the saving behaviour: that optimal saving may have to be positive above some threshold $\bar{b}$ (while it can be negative below $\bar{b}$). The reason that the ratio $\bar{c}\,(b)\,/\underline{c}\,(b)$ may be bounded away from one in problem (BL-H) is due to the role of health spending in (BL-L).

The intuition is as follows: If health spending is a luxury good, its share in total expenditures must increase with wealth compared to that of consumption. In other words, in the presence of this luxury good, the marginal value of saving remains high in the low state compared to that in the high state, even as individual becomes wealthier. As such, the precautionary saving motive does not vanish with wealth, as is the case in the income fluctuation problem in (BAH).

To formally derive the conditions under which inequality (5) holds, at least above some wealth threshold $\bar{b}$, we start by characterizing the optimal medical spending in the low-state, in (BL-L).

---

[21] as denoted in Huggett (1993), one can write This claim, stated by Huggett (1993) in Theorem 2, enables him to show that an individual's wealth does not exceed $\bar{b}$. As such, the state-space is bounded and a stationary distribution of wealth can emerge as a result. We postpone the proof of this claim for problem (BAH)—which is different from Huggett (1993)'s proof—to appendix C.2.

When interior, the optimal health spending is given by the following first order condition:

$$\rho \cdot \chi_m (m) \cdot \underline{V} (b') = u_c (c) .$$

Replacing the utility and health production functions by their functional forms, we can rearrange this equation to find optimal health spending as a function of optimal consumption in the low state. This is given by

$$\underline{m} (b) = \left[\frac{\rho \cdot \beta \cdot \underline{V} (\underline{b}' (b))}{\alpha^{\beta}} \right]^{\left(\frac{1}{1+\beta}\right)} \cdot [\underline{c} (b)]^{\left(\frac{\sigma}{1+\beta}\right)} - \left(\frac{h}{\alpha}\right), \tag{7}$$

when the right-hand side of (7) is positive, and zero otherwise. In this equation, $\underline{m} (\cdot)$, $\underline{c} (\cdot)$ and $\underline{b}' (\cdot)$ are the policy functions for medical spending, consumption and future wealth in the low state, respectively.

We are now in a position to prove the following theorem, the central result of this section:

Theorem 3.2 *Under assumption 3.1, saving rate in problem* (BL-H) *cannot be negative above some threshold* $\bar{b}$*, when* $\sigma > 1 + \beta$*.*

Proof. The value function in (BL-L) is non-decreasing in current wealth, and so are the policy functions $\underline{c}(b)$, $\underline{m}(b)$, and $\underline{b}'(b)$; the proof is given in Appendix C.1. As such, we can use (7) to conclude:

$$\begin{aligned} \underline{m} (b) = \left[\frac{\rho \cdot \beta \cdot \underline{V} (\underline{b}' (b))}{\alpha^{\beta}} \right]^{\left(\frac{1}{1+\beta}\right)} \cdot [\underline{c} (b)]^{\left(\frac{\sigma}{1+\beta}\right)} - \left(\frac{h}{\alpha}\right) \\ \geq \left[\frac{\rho \cdot \beta \cdot \underline{V} (\underline{b})}{\alpha^{\beta}} \right]^{\left(\frac{1}{1+\beta}\right)} \cdot [\underline{c} (b)]^{\left(\frac{\sigma}{1+\beta}\right)} - \left(\frac{h}{\alpha}\right) \\ = A \cdot [\underline{c} (b)]^{\left(\frac{\sigma}{1+\beta}\right)} - \left(\frac{h}{\alpha}\right), \end{aligned} \tag{8}$$

where

$$A := \left[\frac{\rho \cdot \beta \cdot \underline{V} (\underline{b})}{\alpha^{\beta}} \right]^{\left(\frac{1}{1+\beta}\right)} > 0.$$

The claim that $A > 0$ follows from the assumption that, even at $b = \underline{b}$, individual can maintain a sustenance level of consumption, delivering a utility that is strictly greater than zero—*i.e.* the utility upon death. When $\sigma > 1$, given $\underline{c}$, the value of being alive, ν, can be chosen sufficiently large to guarantee that this is the case. Since such a stream of consumption is sustainable as long as the individual lives, value function must remain strictly positive for all b.

Let us assume, for the sake of contradiction, that saving is non-positive above some arbitrary threshold $\tilde{b}$. Then, we must have

$$\bar{c}\,(b) \geq r \cdot b + y, \qquad \forall b \geq \tilde{b}.$$

If we let

$$\gamma := \liminf_{b \to \infty} \frac{\bar{c}\,(b)}{\underline{c}\,(b)},$$

condition (5) implies that

$$\gamma \leq \left[\frac{\rho^{-1}(1+r)^{-1} - \pi}{1 - \pi}\right]^{\frac{1}{\sigma}}, \tag{9}$$

above some threshold of wealth. Therefore, we can assume without any loss of generality that, for all $b \geq \tilde{b}$,

$$\underline{c}\,(b) \geq \gamma \cdot \bar{c}\,(b) = \gamma\,(r \cdot b + y)\,, \tag{10}$$

and—from (8),

$$\underline{m}\,(b) \geq A\,[\gamma \cdot \bar{c}\,(b)]^{\left(\frac{\sigma}{1+\beta}\right)} - \left(\frac{h}{\alpha}\right) \geq A\,[\gamma\,(r \cdot b + y)]^{\left(\frac{\sigma}{1+\beta}\right)} - \left(\frac{h}{\alpha}\right). \tag{11}$$

If we replace these in individual's budget constraint, we get:

$$\begin{aligned} \underline{b}'\,(b) &= (1+r)\,b + y - \underline{c}\,(b) - \underline{m}\,(b) \\ &\leq (1+r)\,b + y + \left(\frac{h}{\alpha}\right) - \gamma\,(r \cdot b + y) - A\,[\gamma\,(r \cdot b + y)]^{\left(\frac{\sigma}{1+\beta}\right)} \\ &= -A\,[\gamma\,(r \cdot b + y)]^{\left(\frac{\sigma}{1+\beta}\right)} + (1 + r - \gamma \cdot r)\,b + (1-\gamma)\,y + \left(\frac{h}{\alpha}\right). \end{aligned} \tag{12}$$

When $\sigma > 1 + \beta$, the polynomial on the right-hand side of this inequality is concave and, as such, decreases without bound with b. As a result, the borrowing constraint must bind for all $b \geq \hat{b}$, for some large enough $\hat{b}$. We let $\bar{b}$ denote the maximum of this threshold and $\tilde{b}$:

$$\bar{b} := \max\left\{\tilde{b}, \hat{b}\right\}.$$

The Kuhn-Tucker conditions for problem (BL-H) when the borrowing constraint binds imply that

$$u_c\left(c\right) = \rho \cdot \chi\left(m\right) \cdot \underline{V}_b\left(b'\right) + \mu, \tag{13}$$

where $\mu \geq 0$ is the Lagrange multiplier on the borrowing constraint.[22] Thus, for all $b \geq \bar{b}$, we should have:

$$u_c\left(c\right) = \chi\left(m\right) \cdot \rho \cdot \underline{V}_b\left(\underline{b}\right) + \mu. \tag{14}$$

As noted earlier, the term $\rho \cdot \underline{V}_b\left(\underline{b}\right)$ in this equality is a positive constant, independent of b. Moreover, from (10) and (11), we know that $\underline{c}\left(b\right)$ and $\underline{m}\left(b\right)$ both increase without bound with b. Therefore, $u_c\left(\underline{c}\left(b\right)\right) \searrow 0$ and $\chi\left(\underline{m}\left(b\right)\right) \nearrow 1$, as $b \to \infty$. Because $\mu \geq 0$, equation (14) is therefore bound to be violated for some $b \geq \bar{b}$. This contradicts the optimality of $\underline{c}\left(b\right)$ and $\underline{m}\left(b\right)$, and completes our proof. $\square$

In the proof of theorem 3.2, what causes a contradiction is not whether or not $\bar{c}\left(b\right)/\underline{c}\left(b\right)$ converges to a constant term *per se*. In fact, this ratio can theoretically converge to a constant even when saving remains positive for all b above some threshold $\bar{b}$, as long as this ratio is above the threshold prescribed by (5). What leads to the contradiction is the assumption that saving becomes negative for large enough b. As such, $\bar{c}\left(b\right)$ must increase with $rb+y$ to keep the saving negative. Moreover, negative saving implies that $\bar{c}\left(b\right)$ and $\underline{c}\left(b\right)$ cannot diverge. It is the combination of these two that leads to a contradiction. For saving to remain positive, then, we need the ratio $\bar{c}\left(b\right)/\underline{c}\left(b\right)$ to remain above

$$\left[\frac{\rho^{-1}(1+r)^{-1} - \pi}{1-\pi}\right]^{\frac{1}{\sigma}}.$$

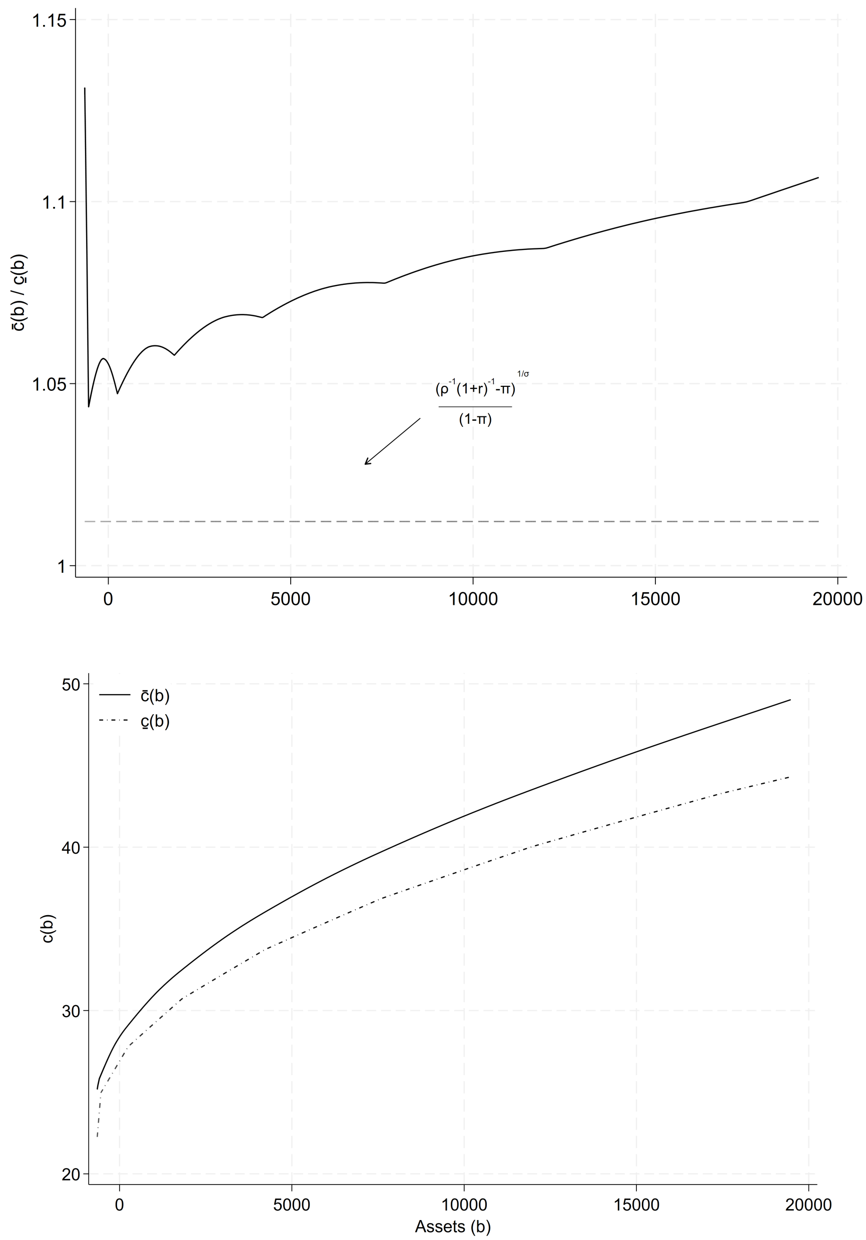
[22] At the threshold $\bar{b}$, the unconstrained choice may just coincide with the borrowing limit, in which case $\mu = 0$. This knife-edge case does not make the problem degenerate or undermine the contradiction. Since $\underline{V}_b(\underline{b}) > 0$, equation (14) with $\mu = 0$ still equates marginal utility to a strictly positive marginal continuation value. As b increases, however, the conjectured allocation implies $u_c(\underline{c}(b)) \to 0$ and $\chi(\underline{m}(b)) \to 1$, whereas $\rho\underline{V}_b(\underline{b})$ remains strictly positive, so the equality must eventually fail. Moreover, beyond the threshold, the upper bound on the unconstrained choice of b' continues to fall below $\underline{b}$; the individual would therefore choose to borrow more than the constraint permits, making the constraint strictly binding and implying $\mu > 0$. The proof itself requires only the weak inequality $\mu \geq 0$.

As Figure 3 demonstrates how, for our sample choice of parameter values, consumption in the high state diverges from total income, leading to increasingly positive saving. Panel (b) in this graph suggests that the ratio $\bar{c}(b)/\underline{c}(b)$ remains consistently above the right-hand side of (5). (In the numerical solutions to the problem in (BL-H) and (BL-L), this ratio seems to be exploding, though at a slow rate, as can be seen in Figure 3 too.)

The condition $\sigma > 1 + \beta$ in theorem 3.2 has an intuitive and straightforward interpretation: As long as the *curvature* of the marginal utility (and not the marginal utility itself) is greater than the *curvature* of the marginal product of health spending (in the health production function), then health is a luxury good. It is important to note that it is not the result that health spending is a luxury that causes precautionary saving to increase without bound, *per se*; but is the fact that such spending is necessitated by an adverse health shock—i.e. endogenous health spending on its own is not the feature that incentivizes continuous saving amounts.

It is also worth mentioning that $\sigma > 1 + \beta$ is not a necessary condition for health spending to increase faster than consumption with wealth, but a sufficient one: Any assumption implied by equation (7) that causes health spending to increase faster than consumption with b suffices for the proof of theorem 3.2 to go through. This can be a high enough value of being alive (ν) or a high enough share of health spending (α) in the health production function, as confirmed by our numerical results.

FIGURE 3. Consumption in Low and High States as Function of Wealth



Notes: For $\bar{c}(b)/\underline{c}(b)$, centred moving averages are drawn to smooth out high-frequency fluctuations at lower wealth levels due to numerical error.

4 Stationary Distribution

In this section, we show how the combined effect of saving at the tail and random separation (i.e., mortality) can lead to a stationary Pareto distribution of wealth in a partial equilibrium setting.

To this end, consider an economy in which, in each period t, a new cohort of individuals—of measure ψ_0—enters the economy with initial wealth distribution $\varphi_0(\cdot)$ (where $\varphi_0(\cdot)$ is a probability measure, defined on (S, S), in which S is a Borel subset of $\mathbb{R}$), in high health.

An individual's problem is the same as before, with healthy individuals facing a probability $1 - \pi$ of transition to the low health state, and individuals in low health with medical spending m facing probability $1 - \chi(m)$ of mortality.

The distribution of wealth in this economy can be written as the sum of the conditional distribution of wealth for individuals in high and low health states, weighted by the relative measure of each. If one of these distributions follows a power law—e.g., is Pareto—then the sum of the two will also follow a power law, even if the other one does not. As noted by Gabaix (2009), *Power Laws in Economics and Finance*:

> "Power laws also have excellent aggregation properties. [...] The general rule is that, when combining two PL variables, the fattest (i.e., the one with the smallest exponent) PL dominates. [...] Hence, multiplying by normal variables, adding nonfat tail noise, or summing over independent and identically distributed (i.i.d.) variables preserves the exponent."

As such, in what follows, we focus on the tail behaviour of the conditional distribution of wealth for high-health individuals, and show that this distribution follows a power law with a power exponent equal to one—i.e., it follows Zipf's law at the tail.

Lemma 4.1 *Any discrete-time, heterogeneous-agent economy with random separation has a stationary state-distribution.*

Proof. Consider an economy in which individuals are heterogeneous in some dimension—say b, where b can be any vector of states. Suppose an initial measure ψ_0 of individuals enter the economy in every period, with initial distribution $\varphi_0(\cdot)$. Of these individuals, fraction $1 - \pi$ separate in each subsequent period.

Assuming P is the economy's transition function and T^* the adjoint operator associated with it, the distribution of individuals who enter in t, in $t+n$ would be

$$T^{*n}\varphi_0(\cdot) = \varphi_n(\cdot).$$

Then, for any $s \in S$, we must have:

$$\left(\frac{\psi_0}{1-\pi}\right)\cdot\varphi(s) = \psi_0\big[\,\varphi_0(s) + \pi\cdot\varphi_1(s) + \pi^2\cdot\varphi_2(s) + \pi^3\cdot\varphi_3(s) + \ldots\big].$$

Since $\varphi_n(\cdot) < 1$ for all n (and $\pi < 1$), this infinite sum is bounded and converges for all s. $\square$

Theorem 4.2 *The stationary distribution of wealth from Lemma 4.1, follows Zipf's law at the upper tail. That is:*

$$\Pr(b \geq x) \;\geq\; \left(\frac{e^{\gamma}\cdot \underline{x}}{x}\right).$$

For some $\gamma > 0$ and $\underline{x} > \underline{b}$.

Proof. Let $\underline{x}$ be some state after which saving is positive (i.e. $b' > b$), and assume $b' = (1+\alpha)\,b$ for all $b \geq \underline{x}$ and some $\alpha > 0$. Because we are only interested in the tail behaviour of the wealth distribution, we assume—without any loss of generality—that $S = [\underline{x}, +\infty)$.

Since a stationary distribution exists, the above-mentioned assumption on saving implies:

$$\Pr(b \geq x) = \pi\cdot\Pr\left(b \geq \frac{x}{1+\alpha}\right).$$

Also, for $\underline{x}$, we must have $\Pr(x \geq \underline{x}) = 1$. If n is the smallest integer for which $x \leq (1+\alpha)^n\cdot\underline{x}$, then

$$\Pr(b \geq x) \geq \pi^n\cdot\Pr(x \geq \underline{x}) = \pi^n.$$

Note that by definition:

$$n = \arg\min_k \left(x - (1+\alpha)^k\cdot\underline{x}\right)^2$$

From the F.O.C we can write n as

$$n = -\frac{\ln(\underline{x}/x)}{\ln(1+\alpha)}$$

we have

$$\ln \Pr(b \geq x) \ \geq \ n \ln(\pi) \ \geq \ -\frac{\ln(\pi)}{\ln(1+\alpha)} \cdot \ln\left(\frac{\underline{x}}{x}\right).$$

Define

$$\gamma = -\frac{\ln(\pi)}{\ln(1+\alpha)} > 0.$$

Thus,

$$\ln \Pr(b \geq x) \ \geq \ \gamma \cdot \ln\left(\frac{\underline{x}}{x}\right),$$

or equivalently,

$$\Pr(b \geq x) \ \geq \ \left(\frac{e^{\gamma} \cdot \underline{x}}{x}\right).$$

This shows that the distribution of wealth in this economy follows Zipf's law in its tail, regardless of what the conditional distribution of wealth for low-health individuals looks like. □

5 Empirical Model & Calibration

We extend the framework by introducing income shocks alongside mortality shocks—as an additional source of heterogeneity—and health insurance. In the empirical model, we have a single dynamic programming problem represeted by FUL. Both low and high-health households face income and health shocks and can insure against it by choosing medical spending.[23] The transition matrix for income is denoted Γ_y and the transition matrix for health is denoted by Γ_h.

[23] The stationary distribution argument stays valid since Pareto Law applies. Intuitively, at the initial date a mass ψ_0 of agents possess high health. In period 1, a fraction transitions to the low–health state and remain there. Within each health group, there are low–income and high–income agents. The analogous stationary distribution argument from the previous section applies here.

Survival is independent of income but depends on health status:

$$\chi(m,h) = 1 - \left(\frac{1}{\alpha m + h}\right)^{\beta}, \qquad \forall\, h \in \mathcal{H}.$$

Unlike the baseline case, a fraction of high–health agents may exit with probability $\chi(m,\bar{h})$. The entry condition ensures that the mass of healthy individuals remains at $\psi_0/(1-\pi)$, implying survival is effectively certain in the high–health state even with minimal medical spending; to ensure this, we set $\bar{h} = 9999$.

We next introduce public health insurance. Once a household moves to the low–health state, it receives a transfer $\underline{m}$. The household's problem is[24]

$$\begin{aligned} V(b,y,h) = \max_{c,m,b'} &\left\{u(c) + \rho\, \chi(\hat{m},h)\, \mathbb{E}_{y',h'}[V(b',y',h')]\right\} \\ \text{s.t.} \quad & b' = (1+r)b + y - c - m, \\ & \hat{m} = \begin{cases} m + \underline{m}, & h = \underline{h}, \\ m, & h = \bar{h}, \end{cases} \\ & b' \geq -\underline{b}. \end{aligned} \tag{FUL}$$

Estimating the Health Production Function

The parameters α and β in the health production function govern how medical spending translates into survival probability. These are estimated by matching model-implied survival to self-reported

[24]Consumption and medical spending are both restricted to be nonnegative, $c \geq 0$ and $m \geq 0$. The nonnegativity of m ensures that the Medicaid transfer $\underline{m}$, which enters only through $\hat{m}$ in the low-health state, cannot be redirected into consumption or saving: a household cannot choose negative medical spending to free up resources elsewhere, so the transfer is effectively earmarked for health spending.

survival expectations and medical spending in the HRS via nonlinear least squares:

$$\min_{\alpha,\beta} \sum_i \left[\chi_i - \left(1 - \frac{1}{(\underline{h} + \alpha\, m_i)^\beta} \right) \right]^2 . \tag{EST}$$

We estimated the health production function using imputed total medical expenditure (Appendix B), which pools information across waves 8–13, yielding $\hat{\alpha} = 0.0136$ and $\hat{\beta} = 0.3928$. For robustness, we also estimated the function using self-reported medical spending from wave 3, obtaining $\hat{\alpha} = 0.0144$ and $\hat{\beta} = 0.345$. The close correspondence between the two sets of estimates suggests that the health production function is stable across data sources and supports the use of the imputed values in our calibration.[25]

Table 1. Nonlinear Least Squares Estimates of Health Production Function

Parameter	Estimate	Robust S.E.	t	p-value	95% C.I.
α	0.0136	0.0044	3.08	0.002	[0.0050, 0.0223]
β	0.3928	0.0009	445.93	0.000	[0.3911, 0.3945]
$N = 4{,}618$, $R^2 = 0.997$, $\underline{h} = 53.2$					

Notes: Nonlinear least squares estimation of $\chi = 1 - (\underline{h} + \alpha m)^{-\beta}$, where χ is self-reported probability of surviving to age 75 and m is imputed total medical expenditure (Appendix B). Sample: single-person HRS households aged 55–64, pooled across waves 8–13 (2006–2016). Robust standard errors.

Table 1 reports the estimation results. Both parameters are precisely estimated. The estimate $\hat{\alpha} = 0.0136$ ($t = 3.08$) confirms that total medical spending has a statistically significant effect on survival, validating the central mechanism of the model. The estimate $\hat{\beta} = 0.3928$ implies that the condition for health spending to act as a luxury good, $\sigma > 1 + \beta$ (Theorem 3.2), is satisfied with

[25] If wealthier households have lower OOP conditional on h due to more generous insurance, $\hat{h}$ may be biased, which could in turn bias $\hat{\alpha}$. Crossley et al. (2022) show that OLS standard errors from regressions with dependent variable understate the true asymptotic variance; applying their correction would yield larger standard errors for $\hat{\alpha}$, though the point estimate remains consistent.

$\sigma = 2$[26]: since $1 + 0.3928 = 1.3928 < 2$, the curvature of marginal utility exceeds the curvature of the marginal product of health spending, ensuring that the precautionary saving motive does not vanish with wealth.

Remaining Parameters

The remaining Calibration targets are $\{\bar{y}, \underline{y}, \Gamma_y, \underline{h}, \Gamma_h, \underline{m}, \underline{b}, \rho, \alpha, \beta\}$, reported in Table 2. The borrowing limit is set to $\underline{b} = -646.7$ thousand dollars, equal to the maximum borrowing observed among HRS households in our sample. Medicaid transfers are calibrated following Kopecky and Koreshkova (2014), who estimate average annual nursing home expenditure for public (Medicaid) payers at \$37,074 per year in 2000 dollars. Since the model period corresponds to two years—matching the HRS's biennial wave structure—the transfer received per period is two years of this annual figure: \$37,074 $\times$ 2 = \$74,148 in 2000 dollars, or \$93,858 in 2010 dollars.[27] That is, $\underline{m} = 93.85$ thousand dollars.

Income states are defined by employment status: the high-income state corresponds to employed households, with mean total household income $\bar{y} = \$87,002$, and the low-income state corresponds to retired households, with mean total household income $\underline{y} = \$41,668$. In HRS data, 293 of 1,717 employed households retire between waves 3 and 4, giving a transition probability of $\gamma_{\bar{y}\to\underline{y}} = 0.17$.

To discipline the probability of entering the absorbing low-health state, we use the spousal loss rate observed in the HRS for couples aged 55–64 between consecutive waves, which is approximately 8%. This treats the death of a spouse as a reduced-form proxy for a large, persistent household-level shock to marginal continuation utility.[28]

$\underline{h}$ is set to match the survival probability when medical spending is negligible. Since zero spending

[26]We adopt $\sigma = 2$ following the calibration strategy of Hosseini (2015).

[27]The conversion uses the CPI index, where the base year is 2010 (CPI = 100). The CPI value for the year 2000 in this index is 79, yielding a conversion factor of $100/79 \approx 1.266$.

[28]De Nardi et al. (2025) jointly model health transitions (including a nursing home category) and widowhood in a couples framework with richer state spaces. They report that the lifetime probability of ever entering a nursing home is 37.2% for women and 26.4% for men in single cohorts aged 70 and older. Assuming roughly 15 years of remaining life, these translate to per-wave entry probabilities of approximately 6% for women and 4% for men.

is not observed in the data, we compute the mean survival probability among households in the bottom one percent of the spending distribution, which yields $\chi = 0.79$. Setting $\underline{h} = 53.2$ in the survival function gives $\chi(0, \underline{h}) = 1 - (1/\underline{h})^{0.3928} = 0.790$, closely matching the data.

Parameters not identified from our data follow Hosseini (2015): borrowing rate 4%, risk aversion $\sigma = 2$, discount factor $\rho = 0.97$. We adopt $\nu = 20$ from Hall and Jones (2007) for the value-of-statistical-life parameter, since the model excludes leisure. Larger values of ν raise upper-tail saving only modestly.

Table 2. Calibration Summary

Parameter	$\bar{y}$	$\underline{y}$	Γ_y	$\bar{h}$	$\underline{h}$	Γ_h	$\underline{m}$	$\underline{b}$	ρ	α	β	r
Value	87.002	41.67	$\begin{bmatrix} 1 & 0 \\ 0.17 & 0.83 \end{bmatrix}$	9999	53.2	$\begin{bmatrix} 1 & 0 \\ 0.08 & 0.92 \end{bmatrix}$	93.85	-646.7	0.97	0.0136	0.3928	0.03
Source	HRS	HRS	HRS	*by const.*	NLS	HRS	K&K (2014)	HRS	Hosseini (2015)	NLS	NLS	Huggett (1993)

Notes: All monetary values in thousands of 2010 dollars. $\bar{y}$ and $\underline{y}$ are mean income for employed and retired households. Γ_y and Γ_h are transition matrices for income and health states. $\underline{m}$ is the average per-period (two-year) Medicaid payment, i.e. two years of the annual long-term care cost estimated in Kopecky and Koreshkova (2014), converted to 2010 dollars. $\underline{b}$ is the borrowing limit, set to the maximum borrowing observed in the HRS sample. α and β are estimated via nonlinear least squares (Table 1). ρ follows Hosseini (2015)."K&K (2014)" refers to Kopecky and Koreshkova (2014). "by const." indicates $\bar{h}$ is set to ensure near-certain survival in the high-health state.

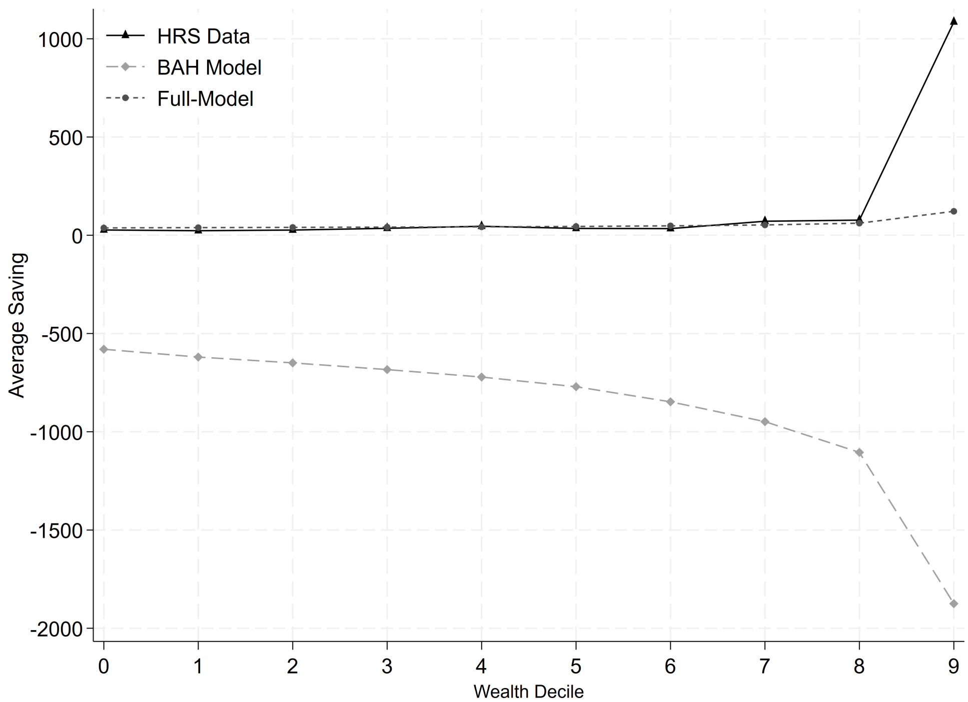
Model Fit

Figure 4 compares model-predicted savings with observed average savings by wealth decile for couple households in the HRS, for both the full model with health shocks and the BAH benchmark without them.

The two models produce starkly different predictions closer to the tail. At deciles 8 and 9, the full model predicts \$52,320 and \$61,180 against \$71,430 and \$77,220 observed, while the BAH model predicts dissaving of −\$948,750 and −\$1,108,260. This comparison indicates that the precautionary mechanism begins to engage at roughly the correct wealth threshold and that the health-shock channel is quantitatively important for reproducing the positive saving observed near the top of the distribution.

At the top decile, the full model predicts saving of \$121,960 against \$1,085,280 observed—a ratio of about 0.11, indicating that the precautionary saving mechanism accounts for approximately 11 percent of observed average saving among the wealthiest couples. The BAH model predicts

Figure 4. Calibration Fit



Notes: Average savings of couple households across wealth deciles (HRS 1996–1998) compared with predictions from the full model and the BAH model. Values in thousands of dollars. The full model captures the acceleration of saving at the top of the wealth distribution, accounting for approximately 11 percent of observed savings in the top decile through the precautionary saving channel alone. The BAH model predicts monotonically increasing dissaving across deciles.

dissaving of $-\$1{,}992{,}990$ at this same decile. The gap between the two model predictions—approximately \$2.1 million—measures the contribution of the health-shock channel to saving at the top of the wealth distribution. This is quantitatively significant given that the full model contains a single novel channel—shocks to marginal continuation value—without invoking bequest motives, return heterogeneity, or entrepreneurial risk.[29]

Figure 4 displays these results visually. The residual gap between the full model and the data at the top decile—approximately \$963,000—is consistent with the presence of additional saving motives that the model deliberately omits, including bequest motives (De Nardi, 2004), return heterogeneity (Benhabib et al., 2015), and entrepreneurial saving (Cagetti and De Nardi, 2006). The contribution of our mechanism is that it identifies a distinct precautionary channel that operates even in the absence of these alternatives.

6 Sensitivity Analysis

This section examines how the model's saving predictions respond to three forces external to the calibration itself: the tightness of the borrowing limit, a technology-driven improvement in health production, and a change in the generosity of the public (Medicaid) health-insurance transfer $\underline{m}$. In each case we re-solve the model holding every other calibrated parameter fixed and report saving by wealth decile alongside the calibrated baseline of Table 2.

Borrowing Limit

The borrowing limit $\underline{b} = -\$646.7$ thousand is calibrated to the most negative net worth observed in the HRS sample—a data-disciplined bound rather than a freely chosen number. Because it is applied uniformly across income states, it is worth checking that the model's saving predictions

[29] The flat saving profile in deciles 1–7 reflects the concavity of the health production function. At low wealth levels, the marginal survival benefit of health spending is still high enough that modest expenditure suffices to balance the value of surviving into the next period. However, as wealth grows, diminishing returns in health production set in—each additional dollar of health spending buys progressively less survival probability. To sustain the marginal gain from health spending relative to the rising value of survival, the household must allocate resources to health at an accelerating rate.

are not an artifact of where this limit happens to be set. Table 3 re-solves the model with the limit halved, to $-\$323.4$ thousand, and again with borrowing eliminated entirely ($\underline{b} = 0$), holding α, β, and $\underline{m}$ at their calibrated values.

TABLE 3. Sensitivity to the Borrowing Limit

Economy	Decile 1	Decile 5	Decile 10	Top-decile share
Baseline ($\underline{b} = -\$646.7\text{k}$)	36.93	42.63	121.96	11.2%
Halved ($\underline{b} = -\$323.4\text{k}$)	37.68 ($+2.0\%$)	43.22 ($+1.4\%$)	122.22 ($+0.2\%$)	11.2%
No borrowing ($\underline{b} = 0$)	38.82 ($+5.1\%$)	44.22 ($+3.7\%$)	122.57 ($+0.5\%$)	11.3%

Notes: All monetary values in thousands of 2010 dollars. Percentages in parentheses are changes relative to the calibrated baseline. Saving computed at each wealth decile's average HRS wealth, interpolated from the solved policy function.

Even a 50 percent tightening of the borrowing limit changes saving by only about 2 percent at the bottom of the distribution and by essentially nothing at the top decile. Eliminating borrowing altogether makes the effect more visible—saving rises everywhere—but the effect remains graded by wealth: largest at the bottom of the distribution (+5.1 percent at decile 1, where households are closest to a binding constraint) and smallest at the top (+0.5 percent at decile 10, which was never close to the constraint regardless of where it is set). This indicates that the borrowing constraint is not what generates the saving pattern documented in this section, particularly at the top of the distribution where the puzzle lies—the precautionary, health-risk channel is doing that work—and that this section's conclusions are not sensitive to the specific calibrated value of the borrowing limit.

Technology-Driven Health Improvements

Like other standard technological progress in the macroeconomics literature, technological improvement in our model also operates through two opposing channels. The degree to which technological improvement contributes to universal healthcare (income effect) and the degree to which newly innovated health products convince consumers to shift their resources further toward health products (substitution effect).

First, note that α relates to the degree of privatization in healthcare: in a welfare-state setting, for example, $\alpha = 0$, meaning that additional private medical spending yields effectively no health increments. However, since private healthcare usually outperforms public healthcare, that case remains an abstract benchmark. Second, β represents the diminishing marginal return to medical spending. In the linear case of the health production function ($\beta = 1$), a unit increment in effective health spending translates into a unit increment in health, regardless of the level of medical expenditure at which this increment occurs. Note that a better health technology that can offer more exotic health services, like advanced regenerative medicine technologies capable of substantially accelerating tissue repair and restoration, can offer very high values of β closer to 1.

Together, these two parameters can be used to study what a technological improvement like AI innovations in the health area implies. The answer depends on two directions. First, if AI research results in new exotic health products and the AI boom is huge enough to push β to twice as high values (for example, previously impossible-to-cure diseases are now offered), given the baseline levels of privatization of the healthcare system, health savings increase among all wealth cohorts since they want to buy those better products in the future. At the same time, if AI is utilized to make previously expensive public healthcare provision cheaper and more universal, and if the gain at the bottom is high, α could significantly decline (to 15 times lower than the baseline in our practice).

Table 4 reports both channels in sequence, holding $\underline{m}$, $\underline{b}$, and all remaining parameters at their calibrated values. Raising β from 0.39 to 0.7 raises saving throughout the distribution, with the top decile rising by about 18 percent and the model's share of observed top-decile saving rising from 11.2 to 12.9 percent. Superimposing a large fall in α (from 0.0136 to 0.0009, representing near-complete socialization of the private return to health spending) reverses this: saving falls by roughly 21 percent relative to the β-only scenario, bringing the model's share of top-decile saving back to essentially the calibrated baseline.

The net effect of technology-driven health improvements is therefore ambiguous in principle, and depends on whether progress mainly raises the productivity of private spending or mainly socializes its return through broader coverage. In most intermediate cases, however, the efficiency channel does not vanish even as coverage expands, since some private margin of spending typically

Table 4. Sensitivity to Technology-Driven Health Improvements

Economy	Decile 1	Decile 5	Decile 10	Top-decile share
Calibrated baseline	36.93	42.63	121.96	11.2%
$\beta : 0.39 \to 0.7$	52.86 (+43.1%)	58.83 (+38.0%)	140.00 (+14.8%)	12.9%
$\hookrightarrow \alpha : 0.0136 \to 0.0009$	32.94 (−37.7%)	38.72 (−34.2%)	119.04 (−15.0%)	11.0%

Notes: All monetary values in thousands of 2010 dollars. Saving computed at each wealth decile's average HRS wealth, interpolated from the solved policy function. The second row raises β alone; the third row additionally lowers α, holding β at 0.7. Top-decile share is model saving divided by observed average top-decile saving ($1,085.28 thousand).

remains; in that sense the precautionary channel identified in this section is more likely to be strengthened than eliminated by continued technological progress.

Health Program Generosity

The next exercise studies the effect of a more generous public health-insurance program. The generosity of health-care coverage has long been a subject of policy debate, with a recurring concern that means-tested programs induce low-wealth households to dissave aggressively in order to qualify for more generous care (Hubbard et al., 1995). To examine this channel in our model, we study how changes in the generosity of the Medicaid transfer, $\underline{m}$, affect saving differently across the wealth distribution.

Our results confirm the usual moral hazard problem in this context (Hubbard et al., 1995; Cutler and Gruber, 1996): a more generous transfer crowds out private precautionary saving, and it does so most where the transfer is largest relative to household resources—at the bottom of the wealth distribution. Doubling $\underline{m}$ from its calibrated value lowers saving by 0.94 percent at the first decile, declining roughly linearly to a 0.49 percent reduction by the ninth decile.

At the top decile, however, the effect nearly vanishes—a reduction of only 0.23 percent, less

TABLE 5. Sensitivity to Medicaid Transfer Generosity

$\underline{m}$	Decile 1	Decile 5	Decile 10	Top-decile share
0 (no transfer)	37.25 ($+0.9\%$)	42.94 ($+0.7\%$)	122.23 ($+0.22\%$)	11.3%
93.85 (calibrated baseline)	36.93	42.63	121.96	11.2%
187.7 (double calibrated)	36.58 (-0.9%)	42.33 (-0.7%)	121.68 (-0.23%)	11.2%

Notes: All monetary values in thousands of 2010 dollars. Percentages in parentheses are changes relative to the calibrated baseline ($\underline{m} = 93.85$). Saving computed at each wealth decile's average HRS wealth, interpolated from the solved policy function.

than half the decline observed one decile below and far below what a linear extrapolation of the bottom-nine-decile pattern would predict. This implies that changes in insurance generosity affect saving behavior asymmetrically across the wealth distribution: households with little wealth to protect respond to a more generous safety net by saving less, while the wealthiest households, whose saving is disciplined primarily by the precautionary motive against catastrophic health spending rather than by eligibility for a fixed transfer, are essentially unaffected.

Risk Aversion

As discussed throughout this section, the key condition that makes health a luxury good is $\sigma > 1 + \beta$ (Theorem 3.2). Since our estimate of β is 0.3928, any value of σ satisfying $\sigma > 1.3928$ ensures that the curvature of the utility function exceeds the curvature of the health production function, so that the precautionary saving motive does not vanish with wealth. We test the robustness of the calibration by moving σ half a point above and half a point below its calibrated value of 2.0.

Raising σ by half a point increases saving throughout the distribution, with lower deciles affected the most: saving at the bottom decile rises by about 36 percent. The wealthiest households, by contrast, increase their saving by only 14 percent, which strengthens our channel further—the

Table 6. Sensitivity to Risk Aversion

Economy	Decile 1	Decile 5	Decile 10	Top-decile share
$\sigma = 1.5$	-4.98 (-113.5%)	-4.38 (-110.3%)	50.78 (-58.4%)	4.7%
Calibrated baseline ($\sigma = 2.0$)	36.93	42.63	121.96	11.2%
$\sigma = 2.5$	50.17 $(+35.9\%)$	56.36 $(+32.2\%)$	139.06 $(+14.0\%)$	12.8%

Notes: All monetary values in thousands of 2010 dollars. Percentages in parentheses are changes relative to the calibrated baseline ($\sigma = 2.0$). Saving computed at each wealth decile's average HRS wealth, interpolated from the solved policy function.

model now accounts for 12.8 percent of saving at the top decile. That is, higher risk aversion affects saving at the bottom of the distribution more than at the top, leaving our channel relatively robust.

Households' response to a symmetric decline in risk aversion is asymmetric. At the top decile, saving declines by 58 percent, and our channel still accounts for 4.7 percent of total saving. This remains a meaningful contribution given how close $\sigma = 1.5$ sits to the 1.3928 threshold, underscoring the robustness of the mechanism.

At the bottom of the distribution, however, the decline is far sharper. The reason is straightforward: at low wealth, the marginal utility of consumption is already high, and the only force sustaining saving is the high marginal return to survival. As σ falls, the marginal utility of consumption declines more and more slowly with wealth, removing the incentive to save.

Statistical Value of Life

Unlike Hall and Jones (2007), our framework does not attempt to explain the statistical value of life or its growth over time. Doing so would require calibrating the model's parameters so that the implied value of life matches empirical estimates of roughly \$2–10 million (2000 dollars).

Instead, because our aim is to study consumer behavior when households self-insure against future mortality shocks through medical spending, what matters is their perceived probability of experiencing such a shock, and hence how much they save to extend their longevity. Consequently, in our framework the value-of-life parameter v and the health-production parameter β can be calibrated separately, unlike studies that require v and β to move together in order to match an empirical value of life. Our estimate of v, the value of life, remains exogenous. Hall and Jones (2007) choose a baseline value of $b = 26$; we use a more conservative estimate of $v = 20$. Table 7 compares our model's response to different calibrated values of v.

Table 7. Sensitivity to the Value of Statistical Life

Economy	Decile 1	Decile 5	Decile 10	Top-decile share
$v = 15$	32.58 (-11.8%)	38.30 (-10.2%)	116.69 (-4.3%)	10.8%
Calibrated baseline ($v = 20$)	36.93	42.63	121.96	11.2%
$v = 25$	39.63 $(+7.3\%)$	45.53 $(+6.8\%)$	125.44 $(+2.9\%)$	11.6%

Notes: All monetary values in thousands of 2010 dollars. Percentages in parentheses are changes relative to the calibrated baseline ($v = 20$). Saving computed at each wealth decile's average HRS wealth, interpolated from the solved policy function.

Raising v from 15 to 25 increases saving throughout the distribution, with the effect graded by wealth: at the bottom decile, moving from $v = 15$ to $v = 25$ swings saving by roughly 19 percentage points relative to baseline (-11.8 to $+7.3$ percent), while at the top decile the swing is much smaller (-4.3 to $+2.9$ percent). This is intuitive: a higher value of life raises the return to precautionary health spending at every wealth level, but the effect is proportionally largest where saving is smallest to begin with.

7 Conclusion

This paper introduces a mechanism through which the precautionary saving motive generates a wealth distribution with a long tail. The mechanism is verified both theoretically and quantitatively. Additionally, it empirically estimates the model parameters and quantifies the gap that the mechanism accounts for compared to models without a persistent precautionary motive.

Theoretically, it shows that, as long as the marginal utility of consumption diminishes faster than the marginal return to survival, incorporating the Hall and Jones (2007) framework into BAH models implies positive saving beyond a wealth threshold, generated entirely by the precautionary saving motive. Second, combining this positive saving behavior with Zipf's law implies the existence of a stationary Pareto distribution above the threshold. This provides a theoretical link between precautionary saving and the emergence of a heavy upper tail in the wealth distribution.

Empirically, it documents that both saving and health spending accelerate among pre-retirement HRS households as wealth rises. It then estimates the parameters of the health production function using nonlinear least squares and finds the health-efficiency parameter $\alpha = 0.0136$, meaning that each additional dollar of health spending yields about 1.36 cents in effective terms. Additionally the estimation process finds that $\beta = 0.39$, which shows how fast health increments diminish with additional increments in effective medical spending αm.

Quantitatively, it finds that the mechanism accounts for approximately 11 percent of total saving among households at the top of the wealth distribution under our baseline calibration. This magnitude is robust to most perturbations, ranging from 11 to 13 percent—in particular, to the borrowing limit, the productivity and degree of privatization of the health-care system, the generosity of the public health-insurance transfer, and the value of life. The exception is the degree to which marginal utility diminishes as consumption rises, which can reduce the explained share of saving at the top to as low as 4.7 percent (when σ is decreased from 2 to 1.5).

The paper can be used in several policy discussions. A sustained improvement in medical technology—of the kind that continued AI-driven progress might plausibly deliver—operates on the model through two channels. First, if better technology makes health spending more productive, households substitute current consumption for medical spending more aggressively, raising saving throughout the distribution. Second, if technological progress is accompanied by broader and better public insurance coverage, the marginal return to a household's private medical spending falls, and it works to reduce saving. The overall effect depends on which channel dominates. Additionally, if AI innovations decline the cost of saving a life, precautionary motive becomes weaker and that further reduces the substitution effect. Additionally, our model predicts that a more generous health programm and better access to the credit market declines the requirement to save for health shocks.

The analysis has several limitations that suggest directions for future research. First, the model is cast in partial equilibrium with exogenous prices. A natural extension is to embed the household problem in a general equilibrium framework to study how endogenous prices affect the precautionary saving motive. In a one-sector model where output can be used for consumption, investment, or health spending, prices equalize and the main GE feedback operates through the interest rate and capital accumulation. Alternatively, a two-sector model with separate production functions for consumption and health goods would allow for endogenous relative prices and sector-specific productivity growth. In that case, an increase in aggregate health spending could raise the relative price of health services, partially offsetting the individual-level incentive to save for future health needs. Whether such GE price effects dampen or amplify the precautionary saving motive is an open question.

Second, the model abstracts from family structure and bequest motives. Households are treated as single decision units, without distinguishing singles from couples or modeling the presence of children. The empirical analysis documents that saving patterns differ substantially between these household types, yet the model does not explain why. Moreover, luxury bequests (De Nardi, 2004) are a well-known alternative mechanism for generating upper-tail wealth concentration. Introducing a bequest motive could interact with the health-spending channel in nontrivial ways: if agents value leaving wealth to heirs, saving at the top of the distribution could rise beyond what the precautionary motive alone implies, complementing the mechanism studied here. Conversely, if bequests and health spending compete for the same resources near the end of life, the two motives could partially offset. Disentangling the relative importance of these mechanisms is an important direction for future work.

A Data Appendix

A.1 Definitions and Measurement

This section defines the saving measures used throughout this section and discusses practical considerations that arise when implementing them in the data. We define three objects: total saving, active saving, and saving rates. Total saving measures the change in household net worth between survey waves. Active saving removes the component of wealth growth attributable to returns on existing assets, isolating the portion that reflects discretionary accumulation. Saving rates scale each measure by net worth, allowing comparisons across households with different wealth levels.

Total Saving

The primary measure of saving is the change in household net worth between consecutive survey waves:

$$S_{it} = b_{it+1} - b_{it} \tag{15}$$

where b_{it} denotes the net worth of household i at the end of period t. Since the HRS interviews households every two years, S_{it} captures wealth accumulation over a two-year horizon.

An alternative derivation proceeds from the household budget constraint. Let r denote the return on wealth, y_{it} labor income, c_{it} consumption, and m_{it} medical and other non-consumption expenditures. The budget constraint implies:

$$b_{it+1} = b_{it} + r\, b_{it} + y_{it} - c_{it} - m_{it} \tag{16}$$

so that saving can equivalently be expressed as the difference between total income and total expenditure:

$$S_{it} = r\, b_{it} + y_{it} - c_{it} - m_{it} \tag{17}$$

Although the two expressions are equivalent in theory, they differ in practice because each relies on different variables measured with different degrees of precision. The income-expenditure definition requires accurate measurement of consumption, income, and returns—each of which is subject to substantial measurement error in survey data (Juster et al., 1999). The first-differencing definition in (15) requires only two observations of net worth. Since the HRS collects detailed asset and liability data at the household level, we adopt (15) as our baseline measure.

Active Saving

Total saving includes the component of wealth growth generated by returns on existing assets. In many cases—particularly for financial wealth held in retirement accounts, index funds, or other managed portfolios—these returns are automatically reinvested without any discretionary action by the household. To isolate the portion of saving that reflects active household decisions, we define active saving as:

$$S_{it}^{act} = S_{it}^{tot} - r_{it}\, b_{it} \tag{18}$$

where $r_{it}\, b_{it}$ denotes capital income. The distinction between total and active saving is informative about the sources of wealth accumulation. If total and active saving rates diverge substantially in a given wealth bracket, this indicates that returns on existing wealth—rather than new saving out of income—account for most of the observed wealth growth. As we document in Section A.3, this divergence is particularly pronounced at the top of the wealth distribution.

Saving Rates

To compare saving intensity across households with different wealth levels, we define the total saving rate and the active saving rate as:

$$s_{it}^{tot} = \frac{S_{it}^{tot}}{|b_{it}|}, \qquad s_{it}^{act} = \frac{S_{it}^{act}}{|b_{it}|} \tag{19}$$

The saving rate directly determines whether someone moves up or down the distribution. Two people with the same absolute saving can have wildly different wealth trajectories if their starting wealth differs — a \$5K saving from \$50K wealth is a 10% rate; from \$500K it's 1%. Only the rate tells you whether wealth is converging or diverging.

Bach et al. (2017) make an explicit decomposition: they split wealth accumulation into a return component and a saving component, both measured as rates, because the multiplicative nature of wealth growth means rates, not levels, drive the long-run shape of the distribution. In this text, Whenever not specified, saving-rates refer to total saving rate rather than active saving rate.

A.2 Data and Sample Selection

This section describes the data (Health and Retirement Study, 2022; RAND Center for the Study of Aging, 2022), explains the cohort and age-group selection, documents the sample restrictions, and benchmarks the wealth distribution in the selected sample.

Health and Retirement Study

The Health and Retirement Study (HRS) is a longitudinal panel survey of Americans over the age of 50, sponsored by the National Institute on Aging (grant number NIA U01AG009740) and conducted by the University of Michigan. Households are interviewed every two years, beginning in 1992, with asset and liability data collected at the household level at the end of each survey year. We use the RAND HRS Longitudinal File, a cleaned and harmonized version of the data that includes derived variables for demographics, income, and wealth components.

The HRS follows several cohorts, each entering the sample at a different date and re-interviewed biennially thereafter. The Survey is designed to be nationally representative of the non-institutionalized US population aged 50 and older. To maintain representativeness of the aging population, new cohorts are added periodically. Since wealth is recorded at the household level, the unit of analysis differs by household type: for singles, the individual and the household coincide; for couples, both spouses share a single wealth observation.

Cohort and Age-Group Selection

The analysis focuses on households in wave 3 (the 1996 interview) with at least one member aged 55 to 64. This choice is motivated by two considerations. First, this age-wave cell contains the largest number of observed couple households—1,553 couples along with 1,568 singles—ensuring adequate sample size for decile-level analysis. Table A1 reports the number of households across age groups and waves for both household types, confirming that the 55–64 group in wave 3 provides the best combination of sample size and demographic homogeneity.

Second, the 55–64 age range captures households in the final decade before typical retirement. These households face active portfolio and saving decisions while simultaneously confronting rising health risk and approaching Social Security eligibility. This makes them well suited for studying the relationship between wealth accumulation and saving behavior that motivates the subsequent sections.

Since the HRS interviews households every two years, saving is measured as the change in net worth between wave 3 (1996) and wave 4 (1998), yielding a two-year saving flow for each household.

Table A1. Number of Households by Age Group and Wave

a. Couples

Wave	55–64	65–74	75–84	85–94
1992	1,468	—	—	—
1994	1,507	741	485	70
1996	**1,553**	582	504	79
1998	1,552	1,171	507	77
2000	1,416	1,124	498	64
2002	1,265	1,142	552	61
2004	1,091	1,120	564	74
2006	880	1,115	573	81
2008	833	1,107	615	93
2010	1,125	1,017	599	89
2012	1,213	917	612	114
2014	1,176	791	570	107
2016	1,170	664	542	106
2018	944	613	510	94
Total	17,193	12,104	7,131	1,109

b. Singles

Wave	55–64	65–74	75–84	85–94
1992	1,278	—	—	—
1994	1,432	1,031	1,839	676
1996	**1,568**	690	1,860	869
1998	1,370	1,332	1,764	988
2000	1,111	1,258	1,517	964
2002	922	1,264	1,309	929
2004	840	1,188	1,061	838
2006	759	1,176	886	775
2008	800	1,009	800	714
2010	1,612	803	813	621
2012	1,762	662	785	475
2014	1,764	678	756	392
2016	1,993	804	713	343
2018	1,778	822	535	288
Total	18,989	12,717	14,638	8,872

Notes: This table reports the number of households by age group and interview wave. Panel (a) shows couple households; panel (b) shows single-person households. The selected analysis sample—wave 3, ages 55–64—is indicated in bold. Cells marked "—" indicate that the cohort had not yet entered the sample.

Sample Restrictions

We impose several restrictions on the analysis sample. First, we require that saving rates be non-missing. Since the saving rate is defined as saving divided by net worth, this restriction excludes two types of households: those that attrit between waves 3 and 4 (and therefore lack the wealth observation needed to compute saving), and those with exactly zero net worth (for which the saving rate is undefined). For couple households, these conditions must hold for both spouses; if either spouse has a missing saving rate, the entire household is dropped. This restriction removes 181 couple households and 375 single households.

Second, we exclude households with extreme saving rates. Observations with saving rates below the 1st percentile or above the 99th percentile of the saving rate distribution are dropped. For couple households, if either spouse's saving rate falls in the extreme tails, both spouses are excluded. This trimming removes an additional 26 couple households and 22 single households, limiting the influence of outliers arising from large capital gains or losses, inheritances, or measurement error in asset valuations. Households with negative net worth are retained throughout. For these observations, the saving rate measures the share of debt repaid between waves. After all restrictions, the analysis sample comprises 1,346 couple households and 1,171 single-person households.

Saving Rates

The normalization introduces several measurement issues. First, households with zero net worth produce an undefined saving rate. We exclude these observations: 56 couple households and 539 single households. Second, households with negative net worth are retained. For these households, the saving rate measures the share of debt repaid; for example, a household that reduces its net liabilities from $100,000 to $50,000 has a saving rate of one-half. Third, because the HRS measures wealth every two years, the saving rate reflects a two-year accumulation rate rather than an annual flow. We report all saving rates on a per-wave (two-year) basis throughout this section.

A further consideration is the treatment of outliers. Both saving levels and saving rates exhibit extreme values at the tails of the wealth distribution, driven by large capital gains or losses, inheritances, or measurement error in asset valuations. Rather than trimming observations ex ante, we report robust measures of dispersion—the interdecile range alongside the standard deviation—and quantile-based skewness to characterize the shape of the distribution without being dominated by extreme observations. This approach follows the methodology used in recent studies of saving behavior using administrative data (Bach et al., 2017; Fagereng et al., 2019).

Wealth Concentration in the Sample

Figure 5 displays the share of total wealth held by each wealth decile in our sample, separately for couples and singles. The wealth distribution is heavily concentrated at the top: among singles, the top decile holds 63.2 percent of total wealth, while among couples the corresponding share is 52.2 percent. This degree of concentration is broadly consistent with the patterns documented in aggregate U.S. wealth data (De Nardi and Fella, 2017), where the top 10 percent of the distribution typically accounts for the majority of total net worth.[30]

The cross-sectional wealth distribution in the sample is broadly consistent with the patterns documented in the quantitative wealth-distribution literature (DeNardi and Fella, 2017; Saez and Stantcheva, 2016). As expected, households in the lower deciles hold a small fraction of total wealth, while the top decile accounts for a large share; Tables A2 and A3 report these distributional properties and confirm that the descriptive moments align with US aggregate estimates.

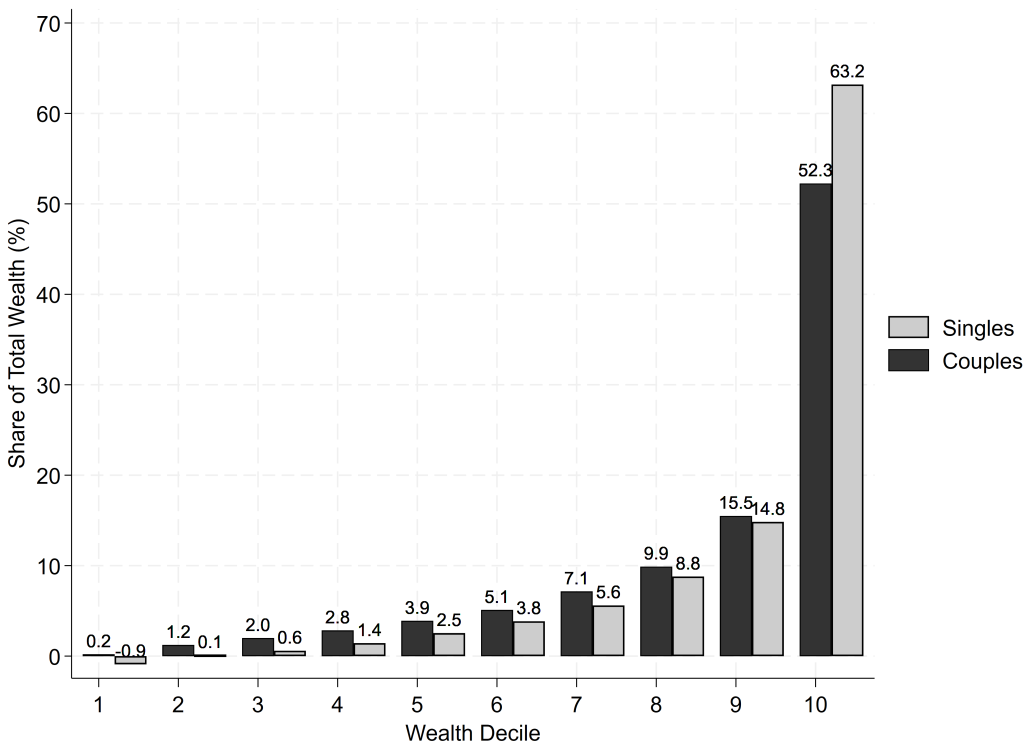
This section does not attempt to contribute to the wealth-distribution literature, which is by now extensive (Guvenen et al., 2015; Cagetti and De Nardi, 2008). Instead, our focus is on saving behaviour, and we verify that the saving patterns we document are robust to the use of aggregate US data (Saez and Zucman, 2016).

A.3 Results

This section presents the main empirical findings. It begins with the distribution and composition of wealth, then examines saving levels and saving rates across wealth deciles. It characterizes the dispersion and symmetry of saving rates and compares these patterns with those documented in

[30]The distributional properties we find here in general are inline with

Figure 5. Share of Total Wealth by Decile



Notes: This figure displays the share of total wealth held by each wealth decile among households aged 55–64 in wave 3 (1996). Red bars indicate single-person households; black bars indicate couples. The share is computed as the sum of net worth within each decile divided by the total net worth of the respective household type.

administrative data from other countries.[31] Finally, it examines the correlation between saving rates and different components of income, highlighting the role of wealth returns.

Throughout, we analyze couple and single households separately. As discussed in Section A.1, wealth is measured at the household level, so the two groups are not directly comparable in levels. The focus is on how saving behavior varies across the wealth distribution within each group.

Wealth Distribution and Composition

Table A2 reports the mean and median net worth for each wealth decile, separately for singles and couples. For singles, the mean and median are closely aligned in the middle of the distribution, indicating relatively little within-decile skewness. The exceptions are the tails: the bottom decile has a mean net liability of $21,000 against a median of $3,000, reflecting a small number of heavily indebted households, while the top decile has a mean of $1,442,000 against a median of $833,000, indicating substantial right skew among the wealthiest singles.

Couples hold substantially more wealth than singles at every point of the distribution. Even after dividing couple wealth by two to approximate a per-capita measure, the gap persists: at the bottom, per-capita couple wealth is $6,000 compared to −$21,000 for singles; at the top, it is $1,535,000 compared to $1,442,000. This difference likely reflects both selection into marriage and the economic advantages of resource pooling. Among couples, within-decile skewness is modest below the ninth decile but pronounced at the top, where the mean-median gap exceeds $1.2 million.

Table A3 reports the composition of wealth across deciles. For singles, housing is the dominant asset from the third through the eighth decile, peaking at 68 percent of net worth in the fifth decile. Financial assets gain importance higher in the distribution, rising from −8 percent in the third decile to 36 percent in the top decile, where they become the largest single component. The bottom two deciles are distinct: the first decile holds wealth almost entirely in financial assets (104 percent of net worth, offset by liabilities in other categories), shile the second decile is concentrated in transportation and financial assets, consistent with households that do not own homes.

[31]This comparison is descriptive: it helps orient the reader to how wealth distribution and life-cycle saving patterns look in other countries.

Table A2. Mean and Median Wealth by Decile (Thousand Dollars)

	Singles		Couples	
Wealth Decile	Mean	Median	Mean	Median
P0–P10	−21	−3	12	19
P10–P20	2	1	71	71
P20–P30	13	13	116	115
P30–P40	32	32	168	168
P40–P50	57	57	225	223
P50–P60	87	87	299	299
P60–P70	127	126	413	410
P70–P80	197	194	581	568
P80–P90	336	332	903	871
P90–P100	1,442	833	3,070	1,820

Notes: Net worth at the beginning of wave 3 (1996), in thousands of real dollars. The gap between mean and median within a decile indicates within-decile skewness.

TABLE A3. Wealth Composition by Decile (Percent of Net Worth)

a. Singles

Wealth Decile	Housing	Financial	Transport.	Pension	Real Estate	Business
P0–P10	1	104	−4	−1	0	0
P10–P20	2	32	67	0	0	0
P20–P30	52	−8	43	7	5	1
P30–P40	64	8	16	7	4	1
P40–P50	68	10	13	4	4	1
P50–P60	56	15	8	8	10	2
P60–P70	60	14	9	8	9	1
P70–P80	51	21	6	11	8	2
P80–P90	38	28	4	13	14	3
P90–P100	23	36	2	13	19	6
Total	42	26	16	7	7	2

b. Couples

Wealth Decile	Housing	Financial	Transport.	Pension	Real Estate	Business
P0–P10	39	32	32	1	−3	−1
P10–P20	69	3	16	6	4	2
P20–P30	68	9	13	3	5	1
P30–P40	53	15	11	9	9	3
P40–P50	49	14	11	13	9	3
P50–P60	43	21	9	14	9	5
P60–P70	34	22	7	18	15	4
P70–P80	28	27	5	19	14	7
P80–P90	23	27	4	19	19	9
P90–P100	14	27	2	12	27	17
Total	42	20	11	11	11	5

Notes: Each cell reports the average share of net worth held in the given asset category, computed at the household level and averaged within each decile. Housing refers to the net value of the primary residence. Financial includes stocks, mutual funds, money market accounts, CDs, bonds, and other financial assets. Transport. refers to vehicles, boats, and similar assets. Pension denotes IRA and Keogh accounts. Real Estate is the net value of non-primary properties. Business is the net value of owned enterprises. Shares can exceed 100% or be negative when a household holds positive assets in one category but has negative net worth overall.

The composition of couple wealth follows a similar gradient but with greater diversification at higher wealth levels. Housing dominates for lower deciles, peaking at 69 percent in the second decile. As wealth increases, the portfolio shifts toward financial assets, pensions, real estate, and business equity, which collectively account for 83 percent of net worth in the top decile. Notably, business equity rises from under 3 percent in the bottom half to 17 percent at the top—a pattern largely absent among singles. Across all households, housing accounts for 42 percent of total wealth for both singles and couples. The key difference is that singles concentrate the remainder in financial and transportation assets, while couples hold a more diversified portfolio across pensions, real estate, and businesses.

This compositional shift is important for interpreting the saving results that follow. As documented below, wealth returns are more strongly correlated with saving at the top of the distribution, precisely where portfolios tilt toward return-generating assets such as financial holdings, real estate, and business equity.

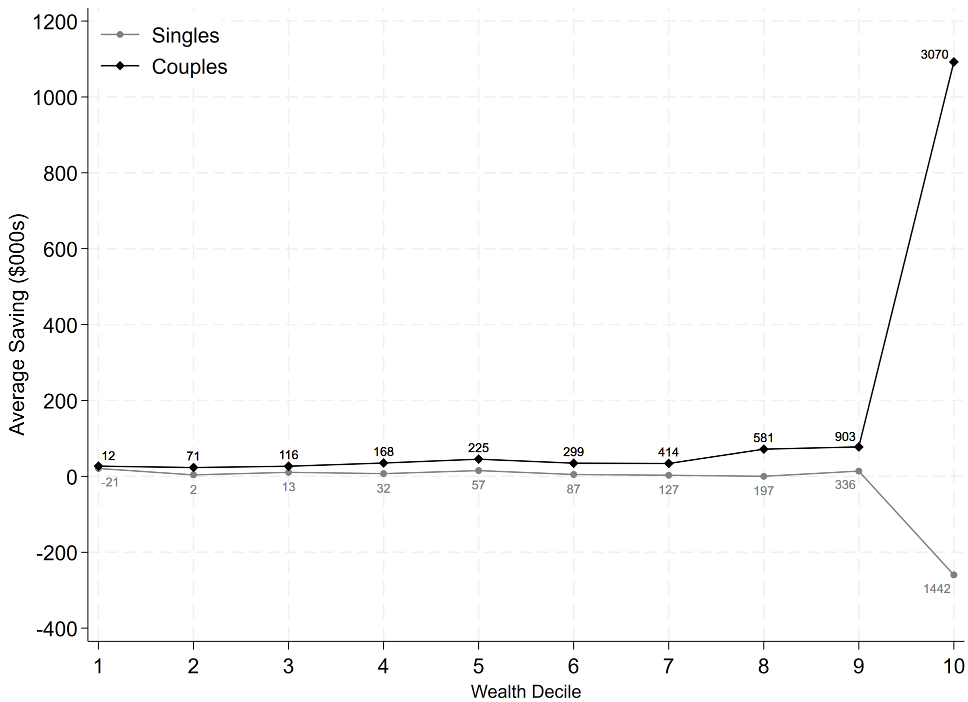
Saving Levels

Figure 6 presents average saving—the change in net worth between waves 3 and 4—by wealth decile for both household types. The patterns differ sharply. Among couples, saving is positive throughout the distribution, ranging from $23,000 to $45,000 in the bottom seven deciles, rising to $71,000 and $78,000 in the eighth and ninth, and reaching $1,092,000 in the top decile. The increase is monotonic and accelerates at the top.

Among singles, the pattern is different. The bottom decile saves $21,000 on average, and the middle deciles save modest amounts between zero and $15,000. The top decile, however, dissaves $260,000 on average. This dissaving at the top is driven by large negative outliers—a feature we return to in the dispersion analysis below.

The divergence between couples and singles at the top of the wealth distribution is striking. In the highest decile, couples save $1,092,000 while singles dissave $260,000—a gap of $1,352,000. For couples, the pattern is inconsistent with standard life-cycle models, which predict that wealthy households should draw down assets to finance consumption. While consumption data are not observed in the HRS, the magnitude of couple saving at the top suggests that these households maintain high consumption levels while simultaneously accumulating wealth—a pattern that

FIGURE 6. Average Saving by Wealth Decile



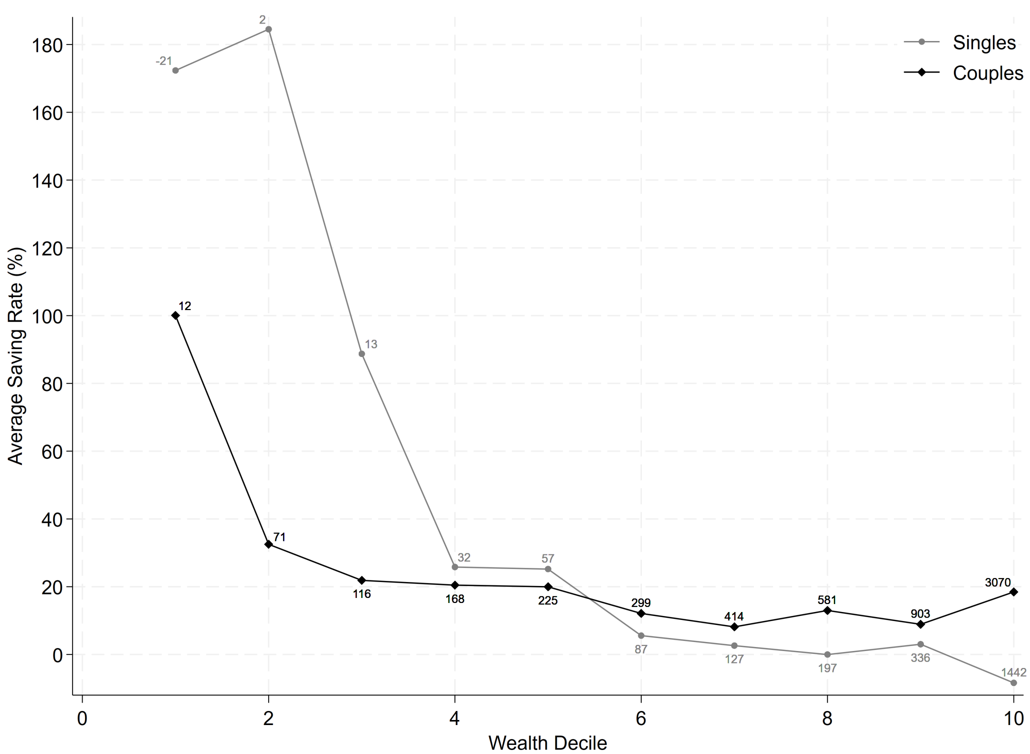
Notes: This figure reports average saving (change in net worth from wave 3 to wave 4, in thousands of real dollars) by wealth decile. Gray circles indicate single-person households; black diamonds indicate couples. Wealth deciles are constructed separately for each household type. Average wealth in each decile is reported by numbers near the bullets for singles and near the diamonds for couples.

requires explanation beyond standard consumption-smoothing motives.

Saving Rates

To compare saving intensity across households with different wealth levels, Figure 7 reports saving rates—saving divided by net worth—by wealth decile. Among singles, the bottom two deciles exhibit elevated saving rates (172 and 185 percent), reflecting the small denominators in these groups. From the third decile onward, the saving rate declines steadily, reaching approximately zero by the eighth decile and turning negative (−8 percent) in the top decile.

FIGURE 7. Average Saving Rate by Wealth Decile



Notes: This figure reports the average saving rate (saving divided by net worth) by wealth decile. Gray circles indicate singles; black diamonds indicate couples. Saving rates are computed at the household level over a two-year horizon (wave 3 to wave 4). Average wealth in each decile is reported by numbers near the bullets for singles and near the diamonds for couples.

For couples, the saving rate in the bottom decile is 99 percent, declining to 32 percent in the second and stabilizing around 20 percent in deciles three through five. It falls further to approximately 10 percent in deciles six through nine, then rises to 20 percent in the top decile. This U-shaped pattern at the top—where saving rates increase rather than continue declining—is of central importance for upcoming sections and this section is the first to record it for the HRS dataset. Standard income-fluctuation models predict that saving rates become negative as wealth increases, since wealthier households can smooth consumption without additional accumulation. The data contradict this prediction for couples.

Table A4 decomposes the saving rate into contributions from each asset category. For singles, saving in the bottom deciles is driven by gains in housing, financial, and transportation assets. In the upper deciles, the pattern reverses: housing and real estate values decline as a share of net worth, while financial saving provides the only positive contribution—except in the top decile, where all major categories show declines.

For couples, financial and pension saving are the primary sources of wealth accumulation across most of the distribution. Unlike singles, couples do not exhibit dissaving in housing at the top. Business equity begins to contribute meaningfully from the seventh decile onward. Overall, couples tend to save across multiple asset classes rather than drawing down in one to fund accumulation in another—a pattern consistent with their more diversified portfolios.

Figure 8 compares total and active saving rates across wealth deciles. Active saving—which subtracts capital income from total saving—isolates the discretionary component of wealth accumulation. The two measures track each other closely in the lower deciles but diverge at the top, particularly for couples. In the top couple decile, the active saving rate is approximately 10 percentage points below the total rate, indicating that roughly half of observed saving at the top is attributable to returns on existing wealth rather than new accumulation out of income.

For singles, the gap between total and active saving is largest in the bottom two deciles—approximately 50 and 78 percentage points—consistent with the finding that financial saving drives wealth growth at the bottom. At the top, the two measures converge, reflecting the negative total saving rate in the top single decile.

The divergence between total and active saving rates at the top of the couple distribution underscores the importance of wealth returns for understanding saving behavior among the wealthy.

Table A4. Saving Composition by Decile (Percent of Net Worth)

a. Singles

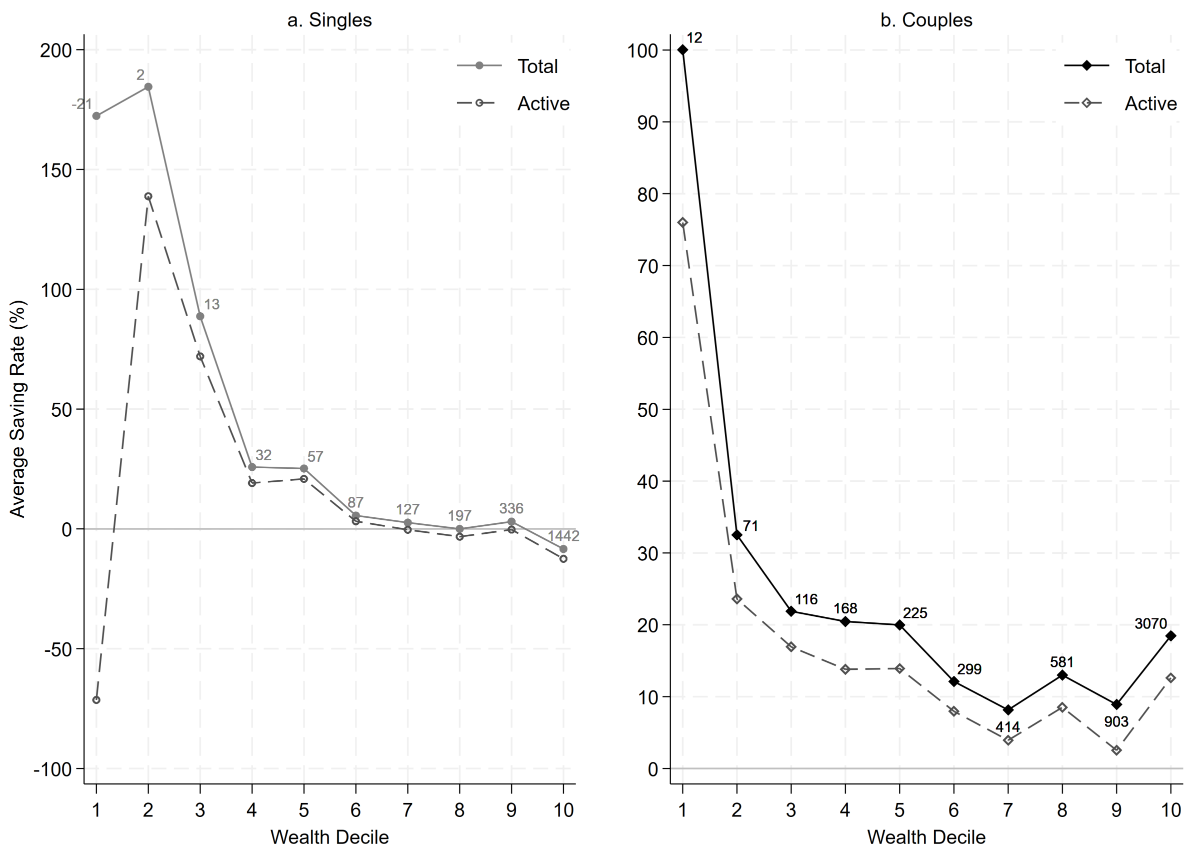
Wealth Decile	Housing	Financial	Transport.	Pension	Real Est.	Business	Total
P0–P10	37	87	42	6	0	0	172
P10–P20	50	50	84	0	0	0	185
P20–P30	76	14	−2	0	−3	4	89
P30–P40	13	1	−2	2	13	−1	26
P40–P50	18	4	2	1	−1	1	25
P50–P60	−1	4	2	0	−2	3	6
P60–P70	0	3	−1	7	−5	0	4
P70–P80	−3	4	0	1	−2	0	−1
P80–P90	−2	5	1	1	−5	3	3
P90–P100	−2	−4	1	1	−3	−1	−8
Total	19	17	13	2	−1	1	50

b. Couples

Wealth Decile	Housing	Financial	Transport.	Pension	Real Est.	Business	Total
P0–P10	42	40	18	1	0	−1	100
P10–P20	7	11	1	11	4	0	32
P20–P30	1	9	2	3	2	4	22
P30–P40	5	8	1	4	1	−1	20
P40–P50	1	16	−1	8	−3	0	20
P50–P60	2	5	0	8	−2	0	12
P60–P70	0	4	0	6	−3	1	8
P70–P80	0	6	0	6	−2	4	13
P80–P90	0	3	1	7	−2	1	9
P90–P100	3	10	0	3	−1	3	18
Total	6	11	2	6	−1	1	26

Notes: Each cell reports the average ratio of saving in the given asset category to total net worth, computed at the household level and averaged within each decile. The Total column equals the sum of all components and corresponds to the total saving rate. All figures are in percent. Average wealth in each decile is reported by numbers near the bullets for singles and near the diamonds for couples. Average wealth in each decile is reported by numbers near the bullets for singles and near the diamonds for couples

Figure 8. Total vs. Active Saving Rates by Wealth Decile



Notes: This figure compares total saving rates (solid lines) with active saving rates (dashed lines) by wealth decile. Active saving subtracts capital income from total saving. Panel (a) shows singles; panel (b) shows couples. The horizontal line at zero is included for reference.

We examine this relationship more formally in Section A.3.

Dispersion and Symmetry

Table A5 reports three measures of the saving rate distribution within each wealth decile: the standard deviation, the interdecile range (the difference between the 90th and 10th percentiles), and quantile-based skewness.[32] The interdecile range provides a robust complement to the standard deviation: in a standard normal distribution, the ratio of the interdecile range to the standard deviation is approximately 2.56, so ratios substantially below this threshold suggest the presence of heavy tails or outliers that inflate the standard deviation relative to the interquartile spread.

Table A5. Dispersion and Symmetry of Saving Rates

	Standard Deviation		Interdecile Range		Skewness	
	Singles	Couples	Singles	Couples	Singles	Couples
P0–P10	430%	174%	641%	448%	0.38	0.47
P10–P20	610	104	886	176	0.80	0.35
P20–P30	239	68	521	133	0.55	0.19
P30–P40	258	67	238	139	0.17	0.28
P40–P50	112	81	192	139	0.30	0.22
P50–P60	76	54	171	111	−0.02	0.16
P60–P70	57	55	134	124	0.03	0.29
P70–P80	62	67	132	117	0.17	0.19
P80–P90	79	48	155	111	0.22	0.12
P90–P100	52	97	136	150	0.23	0.26
Total	198	82	320	165	0.28	0.25

Notes: Standard deviation and interdecile range (P90 − P10) are expressed in percentage points. Skewness is quantile-based: $(P_{90} + P_{10} - 2 \cdot P_{50})/(P_{90} - P_{10})$.

Two patterns emerge. First, dispersion declines with wealth for both household types, but much

[32] Quantile-based skewness is defined as $(P_{90} + P_{10} - 2 \cdot P_{50})/(P_{90} - P_{10})$, where P_k denotes the kth percentile. This measure is bounded between -1 and 1 and is robust to outliers.

more steeply for singles. The standard deviation of the single saving rate falls from 430 percent in the bottom decile to 52 percent at the top—an eightfold reduction. For couples, the decline is more moderate, from 174 to approximately 50–70 percent across most upper deciles. Notably, the couple standard deviation nearly doubles in the top decile (97 percent versus 48 percent in the ninth), a jump not mirrored in the interdecile range (150 versus 111).

Second, saving rates are right-skewed in nearly all deciles for both household types. The positive skewness means that mean saving rates exceed median saving rates. This is visible in the kernel density estimates for the top decile (Figure 9), where the bulk of the distribution lies below zero for both singles and couples, but the couple distribution exhibits several distinct bumps in the right tail that pull the mean upward.

Figure 10 compares the overall saving rate distribution against a normal distribution with the same mean and standard deviation, separately for singles and couples. Both distributions exhibit a narrower central mass and heavier tails than the normal benchmark. The effect is more pronounced for singles, whose distribution has a sharper peak and more extreme outliers. Couples display higher positive saving rates overall, yet their right tail is less extreme than that of singles. In both cases, the right skewness documented in Table A5 is clearly visible.

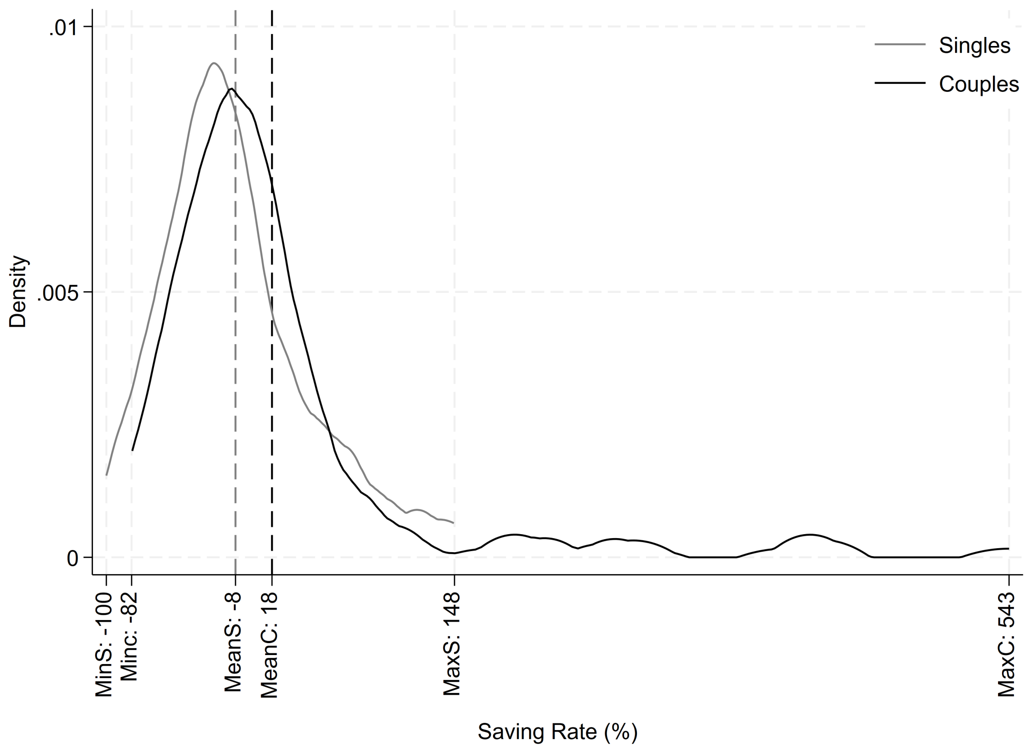
These distributional features—declining dispersion, persistent right skewness, and heavy tails—are consistent with findings from administrative data in other countries. Bach et al. (2017) document a similar pattern in Swedish administrative records, where saving rates at the top of the wealth distribution exhibit high dispersion and positive skewness. Fagereng et al. (2019) report comparable results using Norwegian tax data. The alignment between our HRS-based findings and these administrative studies, which do not suffer from the survey measurement issues inherent in the HRS, supports the robustness of the patterns documented here.

The Role of Wealth Returns

Table A6 reports the correlation between saving rates and three components of income—total income, earnings, and capital income—computed within each wealth decile.[33]

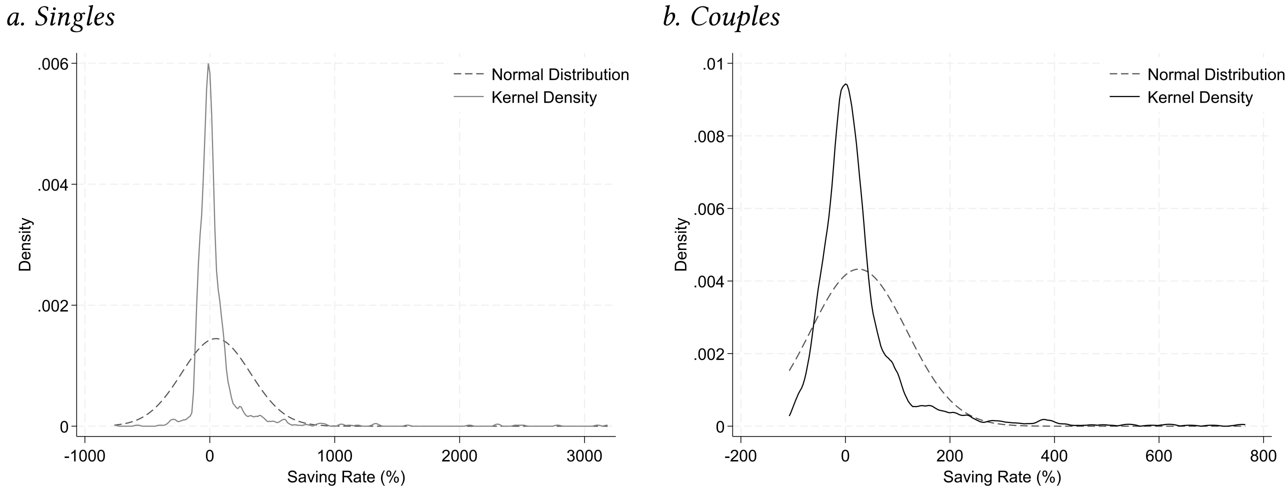
[33] All income variables enter in logs. For households with zero earnings, the log is replaced with the sample minimum of the log-transformed variable.

FIGURE 9. Kernel Density of Saving Rates — Top Wealth Decile



Notes: Kernel density estimates of saving rates in the top wealth decile using the Epanechnikov kernel. Gray denotes singles; black denotes couples. Dashed vertical lines indicate the mean saving rate for each group. MinS, MeanS, and MaxS represent minimum, mean, and maximum of the distribution for singles. MinC, MeanC, and MaxC represent the same statistics for couples.

Figure 10. Kernel Density of Saving Rates vs. Normal Distribution



Notes: Kernel density estimates of saving rates (solid) compared to a normal distribution with the same mean and standard deviation (dashed). Panel (a) shows singles; panel (b) shows couples. Epanechnikov kernel. Note that the range of horizontal axis is different for singles and couples.

Table A6. Correlation of Saving Rates with Income Components

	Total Income		Earnings		Capital Income	
	Singles	Couples	Singles	Couples	Singles	Couples
P0–P10	−0.01	0.14	−0.08	0.16	0.20	0.08
P10–P20	0.11	0.16	0.13	0.09	0.23	0.35
P20–P30	0.01	0.19	0.10	0.22	0.02	0.18
P30–P40	0.17	0.29	−0.03	0.13	0.28	0.34
P40–P50	0.10	0.25	0.10	0.11	0.08	0.31
P50–P60	0.30	0.08	0.16	0.00	0.34	0.19
P60–P70	0.12	0.21	0.07	0.06	0.12	0.17
P70–P80	0.27	0.23	0.15	0.09	0.30	0.20
P80–P90	0.29	0.25	−0.10	0.00	0.30	0.41
P90–P100	0.24	0.34	−0.01	0.10	0.25	0.28

Notes: Pairwise correlations between the saving rate and the natural log of each income component, computed within each wealth decile separately for singles and couples.

The key finding is the contrast between earnings and capital income. The correlation between saving rates and earnings is weak and flat: it remains below 0.2 in nearly all deciles for both household types and shows no systematic pattern across the wealth distribution. Earnings levels do not appear to drive variation in saving behavior within any wealth bracket.

Capital income tells a different story. For couples, the correlation with capital income rises from 0.08 in the bottom decile to 0.41 in the ninth decile, with the top three deciles all exceeding 0.28. For singles, a similar upward pattern is present, with the correlation reaching 0.30 in the eighth and ninth deciles. The total income correlation mirrors the capital income pattern rather than the earnings pattern, confirming that capital income is the dominant component driving the income-saving relationship at higher wealth levels.

These results are consistent with the portfolio composition documented in Table A3: as household wealth increases, the portfolio shifts toward return-generating assets (financial holdings, real estate, business equity), and the saving rate becomes more closely tied to the returns on those assets. This finding is inline with previous research (Bach et al., 2017; Fagereng et al., 2019) that wealth returns—rather than labor income—are the primary driver of saving heterogeneity at the top of the wealth distribution.

A.4 Conclusion

This section documented the distribution of wealth, saving, and saving rates among U.S. households aged 55–64, using the Health and Retirement Study. The analysis revealed several patterns that inform the theoretical and empirical work in the subsequent sections.

Wealth is highly concentrated: the top decile holds 63 percent of total wealth among singles and 52 percent among couples. The composition of wealth shifts systematically across the distribution, from housing and transportation at the bottom to financial assets, pensions, real estate, and business equity at the top. Couples hold more diversified portfolios than singles at every wealth level.

Saving behavior differs sharply between household types. For couples, saving is positive throughout the distribution and increases monotonically with wealth, reaching over $1 million in the top decile. Saving rates decline from the bottom through the middle deciles but flatten and rise at the top. This pattern contradicts the prediction of standard income-fluctuation models, in which saving rates converge to zero or become negative as wealth increases. For singles, saving levels decline at the top of the distribution, and saving rates decrease monotonically from the third decile onward.

The saving rate distribution exhibits declining dispersion with wealth, persistent right skewness across nearly all deciles, and heavy tails relative to the normal distribution. These features are consistent with the patterns documented in Swedish and Norwegian administrative data by Bach et al. (2017) and Fagereng et al. (2019), suggesting that the distributional properties observed in the HRS are consistent with previous studies.

Decomposing the correlation between saving rates and income reveals that earnings explain little of the variation in saving across wealth brackets. Capital income, by contrast, is more strongly correlated with saving rates, and this correlation rises in higher deciles—particularly for couples. This pattern aligns with the return heterogeneity documented in the administrative data literature (Fagereng et al., 2019; Bach et al., 2017) and with the compositional shift toward return-generating assets at the top of the wealth distribution.

B Imputing Total Health Expenditure for the Health and Retirement Study

This appendix explains the process ending up in Figure 2. Using the Medical Expenditure Panel Survey (MEPS), in which total medical expenditure is verified directly by healthcare providers, we estimate the relationship between out-of-pocket and total spending and use it to impute total medical expenditure in the Health and Retirement Study (HRS). The HRS contains detailed data on household wealth but records only out-of-pocket medical costs. The imputation bridges this gap, producing a dataset in which both total medical spending and wealth are available for the same households.

In what follows we are planning to achieve two goals. First, utilizing the imputation procedure by Blundell et al. (2004), we impute the health variables for HRS. Second, we show that using the imputed variable as the dependent variable in an OLS regression preserves consistency of the slope coefficients under standard exogeneity assumptions, which we argue are credible given the provider-verified structure of the MEPS data. Finally, We use the imputed measure to document the medical spending–wealth gradient.

B.1 Imputation Procedures

This section applies the method of Blundell et al. (2004) to impute total medical spending.[34] It then states conditions under which the imputations are consistent in mean and variance. Finally, it present the main theorem for using the imputed values as the dependent variable. Under standard classical assumptions, the theorem shows that OLS estimation with the imputed dependent variable remains unbiased.

Consider the relationship between out-of-pocket spending and total medical spending:[35]

$$p_i = D_i\beta + \gamma h_i + \epsilon_i, \qquad i \in \{M,\, H\}. \tag{20}$$

Here p denotes out-of-pocket spending, D is a vector of demographic and other controls, h is total medical spending, and ϵ is an error term. The index $i \in \{M, H\}$ denotes the donor (MEPS) and main (HRS) samples, respectively. Rearranging (20) yields the imputed total spending:

$$\hat{h}_i = \frac{p_i - D_i\hat{\beta}}{\hat{\gamma}}, \tag{21}$$

where $\hat{\beta}$ and $\hat{\gamma}$ are estimated in MEPS and $\hat{\gamma} \neq 0$.

Note that our imputation target, h_i, is an independent variable when we estimate γ and β in the donor dataset. This is an important consideration, since defining $\hat{h}_i$ as the dependent variable would cause estimation bias in the main sample.

To study the large-sample properties of imputed health spending, it is useful to express imputed total medical spending, $\hat{h}_i$, in terms of the observed variables. Substituting (20) into (21) yields the relationship between the true and imputed values:

[34] The appendix discusses alternative imputation approaches relevant to this study.

[35] See the appendix for a complete discussion of alternative imputation methods.

$$\hat{h}_i = D_i \frac{\left(\beta - \hat{\beta}\right)}{\hat{\gamma}} + \frac{\gamma}{\hat{\gamma}} h_i + \frac{\epsilon_i}{\hat{\gamma}} \tag{22}$$

The imputation embeds both measurement error and systematic drift. As a result, using the imputed values in subsequent regressions can introduce bias, and even simple comparisons of means and variances may be misleading. To see this more clearly, define:

$$\bar{x} = \frac{\mathbf{1}_n' x}{n}, \qquad s_x^2 = \frac{(x - \bar{x})'(x - \bar{x})}{n}, \qquad s_{xy} = \frac{(x - \bar{x})'(y - \bar{y})}{n},$$

where $\mathbf{1}_n$ is an $n \times 1$ vector of ones. Let $\bar{x}$ denote the sample mean of the random variable x, let s_x^2 denote the sample variance of the random variable x, and let s_{xy} denote the sample covariance between the random variables x and y.

It is straightforward to show that if $\hat{\beta}$ and $\hat{\gamma}$ are inconsistent, then the imputed variable will be mean- and variance-inconsistent. The following assumptions ensure consistency of $\hat{\beta}$ and $\hat{\gamma}$:

Assumption B.1 *The donor (MEPS) and main (HRS) samples are drawn from the same underlying population with respect to medical spending. Moreover,*

$$\mathbb{E}[\epsilon] = 0, \qquad \mathbb{E}[D'\epsilon] = 0, \qquad \mathbb{E}[h\,\epsilon] = 0.$$

A common concern in survey-to-survey imputation is measurement error. If the donor sample is measured with error, the consistency of $\hat{\gamma}$ and $\hat{\beta}$ is no longer guaranteed, so a simple OLS fit in the donor data need not be consistent. In our setting, however, the design of the MEPS survey limits the scope for substantial measurement error, making these assumptions reasonable. The MEPS Medical Provider Component (MPC) reduces reporting error by collecting billing/record information directly from providers to supplement and verify household reports of charges and payments.

If the HRS and MEPS represent different underlying populations (so their true average medical spending differs), then a "consistent" imputation in the donor sense is not enough. At best, the imputed values in HRS will converge to the MEPS population mean, because the mapping is estimated to match MEPS. But the relevant benchmark for HRS is the HRS population mean. Therefore, even in large samples, the imputed HRS mean can systematically differ from the true

HRS mean—not because the imputation fails statistically, but because it is targeting the wrong population.

Although MEPS and HRS are not drawn from an identical sampling frame, we can make the two samples effectively comparable by restricting both data sets to the same demographic subpopulation and then applying each survey's calibrated person-level weights. After these restrictions the observable covariate distributions—age, sex, race/ethnicity, education, household size, region—are closely aligned; any residual differences are minimized to the extent that the weighting adjustments bring the weighted moments of the two samples into agreement. [36]

Therefore, under Assumption B.1, the donor (MEPS) and main (HRS) samples share the same underlying population moments. Let μ and σ^2 denote the population mean and variance. Then Assumption B.1 implies

$$\mu_{h_{\mathrm{M}}} = \mu_{h_{\mathrm{H}}} = \mu_h, \qquad \sigma^2_{h_{\mathrm{M}}} = \sigma^2_{h_{\mathrm{H}}} = \sigma^2_h.$$

To be concrete, the next theorem proves the usual consistency condition in standard econometrics textbooks.[37]

Theorem B.1 (Consistency of the OLS coefficients) *Let $(p_i, D_i, h_i)_{i=1}^n$ be an i.i.d. sample drawn from a population that satisfies Assumption B.1. Consider the linear regression*

$$p_i = D_i\beta + \gamma\, h_i + \varepsilon_i, \qquad i \in \{M, H\}, \tag{1}$$

and denote by $X_i := \begin{bmatrix} D_i & h_i \end{bmatrix}$ the $(k+1)$-dimensional regressor ($k = \mathrm{col}(D_i)$). The OLS estimator obtained from the donor (MEPS) sample is

$$\hat{\theta}_n := \begin{pmatrix} \hat{\beta} \\ \hat{\gamma} \end{pmatrix} = \left(X'X\right)^{-1} X'p, \qquad X = \begin{bmatrix} X_1 \\ \vdots \\ X_n \end{bmatrix},\ p = (p_1, \ldots, p_n)'. \tag{2}$$

[36] In other words, by limiting the analysis to a demographically homogeneous slice of the data we approximate the condition required by Assumption B.1 that the donor and recipient samples represent the same underlying population well enough that the imputed expenditures inherit the correct mean and variance properties for the HRS sample.

[37] For instance see Wooldridge (2010)

Then, under Assumption B.1,

$$\operatorname*{plim}_{n\to\infty} \hat{\beta} = \beta, \qquad \operatorname*{plim}_{n\to\infty} \hat{\gamma} = \gamma.$$

Proof. Write $X_i = (D_i, h_i)$ and let $\theta := (\beta', \gamma)'$. From (1) we have the population moment condition

$$\mathbb{E}[X_i\, \varepsilon_i] = 0, \qquad \mathbb{E}[\varepsilon_i] = 0. \tag{3}$$

Assumption B.1 guarantees that all second-order moments exist and that $Q := \mathbb{E}[X_i X_i']$ is positive definite (full rank of the regressor matrix).

By the weak law of large numbers,

$$\frac{1}{n} X'X = \frac{1}{n} \sum_{i=1}^{n} X_i X_i' \ \xrightarrow{p}\ Q. \tag{4}$$

Again by the law of large numbers and (3),

$$\frac{1}{n} X'\varepsilon = \frac{1}{n} \sum_{i=1}^{n} X_i \varepsilon_i \ \xrightarrow{p}\ 0. \tag{5}$$

From (2), (4) and (5),

$$\hat{\theta}_n = \left(\tfrac{1}{n} X'X\right)^{-1} \left(\tfrac{1}{n} X'p\right) = \left(\tfrac{1}{n} X'X\right)^{-1} \left(\tfrac{1}{n} X'(X\theta + \varepsilon)\right).$$

Using (4)–(5) and the continuous-mapping theorem,

$$\hat{\theta}_n \xrightarrow{p} Q^{-1}\, Q\, \theta = \theta.$$

Thus each component converges in probability to its true value: $p\mathrm{lim}\, \hat{\beta} = \beta$ and $p\mathrm{lim}\, \hat{\gamma} = \gamma$.

□

Given the consistency of $(\hat{\beta}, \hat{\gamma})$, it follows that $p\mathrm{lim}\ \bar{\hat{h}}_M = p\mathrm{lim}\ \bar{h}_M$. The drift term vanishes as $\hat{\beta}$ converges to β, and the remaining error component converges to zero by the law of large numbers. Finally, the equality uses $\hat{\gamma}/\gamma \to 1$. Formally, Under assumption B.1, equation (22) implies:

$$
\begin{aligned}
&plim\ \overline{\hat{h}}_M = \text{plim}\ \bar{h}_M = \mu_h, \\
&plim\ s^2_{\hat{h}_M} = plim\ s^2_{h_M} + \frac{1}{\gamma^2}\ \text{plim}\ s^2_{\epsilon_M} = \sigma^2_h + \frac{1}{\gamma^2}\sigma^2_{\epsilon_M}.
\end{aligned} \tag{23}
$$

Within the sample, the imputed values $\hat{h}_{MEPS}$ are mean-consistent, i.e., they converge in probability to the true population mean μ_h. However, the sample variance of the imputed values exhibits a persistent positive bias due to the added measurement/imputation noise[38].

Additionally, properties of the imputed variable in the donor sample need not carry over to the out-of-sample imputations. Trivially, if the joint distribution of demographics and out-of-pocket spending is identical across the two samples, then all properties established above would be inherited. We do not impose that assumption. Applying (21) to the MEPS (donor) and HRS (main) samples yields:

$$
\hat{h}_H = \hat{h}_M + \frac{1}{\widehat{\gamma}}\left[(p_H - p_M) - (D_H - D_M)\,\widehat{\beta}\right]. \tag{24}
$$

Therefore, the average gap in imputed total spending across the two samples is driven by differences in out-of-pocket spending and differences in demographics (scaled by $\widehat{\gamma}$). In particular, if mean out-of-pocket spending is higher in HRS than in MEPS, then—holding demographic differences fixed—the imputed average total spending in HRS will exhibit a systematic upward bias.

Additionally, the standard deviation would still be biased as it was in the original case. The size of the bias depends on the distribution of the out-of-pocket spending and demographics in the two samples. Formally:[39]

[38] This asymptotic behavior follows directly from a classical measurement-error structure. The intuition is simple: we have two random variables x and y related by $y = x + u$, where u is white noise independent of x. It then follows that

$$
\sigma^2_y = \sigma^2_x + \sigma^2_u \geq \sigma^2_x
$$

Where the inequality is strict whenever $\sigma^2_u > 0$. In the MEPS imputation context, this classical errors-in-variables structure induces the inflation of the sample variance by the scaled noise term $\sigma^2_\epsilon/\gamma^2$, even though the mean remains unbiased in the limit.

[39] The first component is obtained by applying the mean operator to both sides of (24), then taking probability limits and using $plim\,\bar{\hat{h}}_M = \mu_h$.

$$
\begin{aligned}
plim\, \bar{\hat{h}}_H &= \mu_h + \tfrac{1}{\gamma}\, plim\left(\bar{p}_H - \bar{p}_M\right) - \tfrac{1}{\gamma}\, plim\left(\bar{D}_H - \bar{D}_M\right) \\
plim\, s^2_{\hat{h}_H} &= \sigma^2_h + \tfrac{1}{\gamma^2}\sigma^2_{\epsilon_M} + \tfrac{1}{\gamma^2}\, plim\left[s^2_{\left(p_H - D_H\hat{\beta}\right)} - s^2_{\left(p_M - D_M\hat{\beta}\right)}\right]
\end{aligned}
\tag{25}
$$

Where $\bar{D}_i = \frac{\mathbf{1}'_n \times D_i\beta}{n}$ for $i \in \{H, M\}$.

Trivially, if out-of-pocket spending in the underlying populations of MEPS and HRS is identical, then the imputed mean of total health spending in the HRS sample is consistent.[40] However, the standard deviation remains biased, at best by a constant positive drift.[41]

As shown in the data section, the demographic composition of MEPS and HRS is very similar for the selected age group. Hence, only an adjustment for differences in out-of-pocket spending is required, as formalized in the following theorem.

Theorem B.2 *Define:*

$$\hat{h}_a \;=\; \hat{h}_H \;-\; \frac{1}{\hat{\gamma}}\left(\bar{p}_H - \bar{p}_M\right).$$

If the underlying demographic distributions in MEPS and HRS coincide and B.1 holds, then $\hat{h}_a$ is mean consistent:

$$plim\, \bar{\hat{h}}_a \;=\; \mu_h,$$

For the second component, note that $s^2_{\hat{h}_H} = \frac{s^2_{p_H - D_H\hat{\beta}}}{\hat{\gamma}^2}$ and $s^2_{\hat{h}_M} = \frac{s^2_{p_M - D_M\hat{\beta}}}{\hat{\gamma}^2}$. Taking the difference yields

$$s^2_{\hat{h}_H} - s^2_{\hat{h}_M} = \frac{s^2_{p_H - D_H\hat{\beta}}}{\hat{\gamma}^2} - \frac{s^2_{p_M - D_M\hat{\beta}}}{\hat{\gamma}^2}.$$

Rearrange by moving $s^2_{\hat{h}_M}$ to the right-hand side and apply the $plim$ operator to both sides. Using (23) delivers the second component.

[40]Set $\mu_{p,HRS} = \mu_{p,MEPS} = \mu_p$. Then

$$plim\left(\bar{p}_H - \bar{p}_M\right) = \mu_{p_H} - \mu_{p_M} = \mu_p - \mu_p = 0.$$

Applying the same argument to the demographic variables yields $plim\, \bar{\hat{h}}_H = \mu_h$.

[41]That is,

$$plim\, s^2_{\hat{h}_H} = \sigma^2_h + \frac{1}{\gamma^2}\sigma^2_\epsilon.$$

Moreover, the variance term for $\hat{h}_H$ converges to the population variance up to a constant drift:

$$\text{plim}\, s^2_{\hat{h}_H} \;=\; \sigma^2_h \;+\; \frac{1}{\gamma^2}\sigma^2_{\epsilon_H}.$$

Proof. Starting from (25), the last term on the right-hand side vanishes. Hence,

$$\text{plim}\,\bar{\hat{h}}_H = \mu_h + \frac{1}{\gamma}\,\text{plim}(\bar{p}_H - \bar{p}_M)\,.$$

It follows that

$$\text{plim}\;\bar{\hat{h}}_a = \text{plim}\,\bar{\hat{h}}_H - \frac{1}{\gamma}\,\text{plim}\,(\bar{p}_H - \bar{p}_M) = \mu_h,$$

which establishes mean consistency of $\bar{\hat{h}}_a$.

For the variance, use (20) to write

$$p_i - D_i\beta = \gamma h_i + \epsilon_i.$$

Under mean independence, $\text{plim}\, s_{h_i,\epsilon_i} = 0$. Therefore,

$$\begin{aligned}\text{plim}\left[s^2_{(p_H - D_H\hat{\beta})} - s^2_{(p_M - D_M\hat{\beta})}\right] \;&= \text{plim}\left(\hat{\gamma}^2\,\left(s^2_{h_H} - s^2_{h_M}\right) + s^2_{\epsilon_H} - s^2_{\epsilon_M}\right)\\ &= \sigma^2_{\epsilon_H} - \sigma^2_{\epsilon_M},\end{aligned}$$

where the last equality follows from Assumption B.1.[42] Substituting this expression back into (25) yields the stated variance result. □

Theorem B.2 presents the consistency result. When working with imputed values, fixed and specific correction terms must be incorporated in the analysis of means and variances. We then examine the conditions under which imputed values can be used in other regression settings and establish the consistency of OLS.

[42]In particular, $\text{plim}\; s^2_{h_H} = \text{plim}\; s^2_{h_M} = \sigma^2_h$.

B.2 Incorporating Imputed Variables into OLS Estimation

To illustrate the issue with using imputed values, consider a simplified single-variable version of (20), given by

$$p_i = \beta_0 + \beta_1 h_i + \epsilon_i \tag{26}$$

Where $i \in \{H, M\}$. Therefore imputed variables can simply be defined as:

$$\hat{h}_i = \frac{p_i - \hat{\beta}_0}{\hat{\beta}_1}$$

Therefore, the imputed values can be expressed as a function of the actual total health spending as follows:

$$\hat{h}_i = \frac{\beta_0 - \hat{\beta}_0}{\hat{\beta}_1} + \frac{\beta_1}{\hat{\beta}_1} h_i + \frac{\epsilon_i}{\hat{\beta}_1}.$$

Now suppose we seek to analyze the association between total health spending and wealth by estimating a simple linear regression model. Specifically, we relate individual health expenditures to wealth as follows:

$$h_i = \theta_0 + \theta_1 W_i + e_i, \tag{27}$$

where h_i denotes total health spending for individual i, W_i represents their level of wealth, θ_0 is the intercept, θ_1 measures the marginal effect of wealth on health spending, and e_i captures unobserved factors affecting health expenditures.

In this specification, total health spending is assumed to affect out-of-pocket spending exogenously,[43] that is, $\sigma_{h_i,\epsilon_i} = 0$. Similarly, for consistency of the OLS estimates, we require $\sigma_{W_i,e_i} = 0$. Under this assumption, the slope coefficient is:

$$\theta_1 = \frac{\sigma_{h_i,w_i}}{\sigma^2_{w_i}}$$

[43]As explained in the empirical section, wealth controls ensures this is the case.

Since we use imputed values in practice, we are estimating $plim\,\hat{\theta} = plim\,\frac{s_{\hat{h}_i,w_i}}{s^2_{w_i}}$. Without imputation, this would converge to θ by the law of large numbers. With imputed values, however, assuming $\hat{\beta}$ is consistent, we instead obtain:

$$plim\,\hat{\theta} = \frac{\sigma_{h_i,w_i}}{\sigma^2_{w_i}} + \frac{1}{\beta} \cdot \frac{\sigma_{\epsilon_i,w}}{\sigma^2_{w_i}} = \theta_1 + \frac{1}{\beta} \cdot \frac{\sigma_{\epsilon_i,w_i}}{\sigma^2_{w_i}}$$

The point of this exercise is straightforward. We have shown that using imputed values does not generally guarantee consistency of OLS estimates, which motivates the need to establish conditions under which consistency is preserved. In particular, some form of an exogeneity argument appears necessary when employing any imputation method.

More intuitively, in the ideal case, including every regressor that could plausibly affect total health spending through wealth in both the imputation equation (26) and the main regression (27) would resolve the issue, as this would satisfy the condition $\sigma_{\epsilon_i,w_i} = 0$. Formally:

$$\begin{aligned} p &= h\,\gamma + D\,\beta_1 + \epsilon \\ h &= w\,\alpha + D\,\beta_2 + u \end{aligned} \tag{28}$$

Where p, h, u and ϵ are $n \times 1$ vectors, D is $n \times k$, and β is $k \times 1$.

The first step is to isolate the imputation regression relationship for γ. To this end, define the projection matrix $P_D = D(D'D)^{-1}D'$ and the residual maker $M_D = I - P_D$. The residualized variable is then given by $\ddot{x} = M_D x$, which allows us to rewrite (28) as:

$$\begin{aligned} \ddot{p} &= \gamma\ddot{h} + \ddot{\epsilon} \\ \ddot{h} &= \alpha\ddot{w} + \ddot{u} \end{aligned} \tag{29}$$

The OLS estimator of α is consistent provided that residualized wealth is orthogonal to both residualized error terms, since:

$$plim\ \hat{\alpha} = \alpha + plim\,(\ddot{w}\ddot{w}')^{-1}\,plim\,(\ddot{w}'\ddot{u}) + plim\,(\ddot{w}\ddot{w}')^{-1}\,plim\,(\ddot{w}'\ddot{\epsilon})$$

The key intuition is straightforward: valid imputation of total health spending requires controlling for all potential sources of confounding. In particular, any variable that affects total health spending through out-of-pocket expenditures must be included in the control set, otherwise the consistency argument would not hold.

We assume that, conditional on demographic controls D, wealth affects out-of-pocket spending only through total health expenditure — that is, $\mathbb{E}[\ddot{w}'\ddot{\epsilon}] = 0$. A potential violation is that wealthier households carry more generous insurance, so that w enters ϵ through the out-of-pocket share even at fixed h. This is an empirical concern rather than a failure of the identification strategy itself: the assumption is imposed at the population level, and demographic controls are intended to absorb systematic variation in the OOP share. Where the residual correlation remains a concern, a natural extension is to instrument for wealth using a variable uncorrelated with insurance generosity conditional on D; we leave this for future work.

A second caveat concerns inference. Crossley et al. (2022) show that the usual OLS standard errors from a regression with a BPP-imputed dependent variable understate the true asymptotic variance, and provide a correction. The point estimates reported in this appendix remain consistent, but hypothesis tests based on uncorrected standard errors may over-reject. Applying the Crossley et al. (2022) correction is left for future work.

Under the additional assumption of exogeneity, $\mathbb{E}[\ddot{w}'\ddot{u}] = 0$, the remaining bias term vanishes, implying consistency of the imputation estimator. The following assumption formally summarizes this argument.

Assumption B.2 *Both MEPS and HRS data sets are from the same underlying population, and* (28) *governs the relationship between total health spending, out of pocket health spending, and wealth. Further, in addition to B.1:*

1. $E[h'\epsilon] = E[w'u] = 0$
2. $E[D'\epsilon] = E[D'u] = 0$

The exogeneity assumption on total health expenditure, $\mathbb{E}[h'e] = 0$, guarantees that wealth is uncorrelated with the residuals in the imputation equation, $\mathbb{E}[w'\epsilon] = 0$. This key condition is typically violated in survey data, as self-reported health spending is subject to measurement error — a concern that has been extensively documented in the literature. However, the structure of the MEPS dataset alleviates this concern, since health expenditure reports are verified directly by the health provider, lending credibility to the exogeneity assumption. The following Theorem and its proof summarize this section.

Theorem B.3 *Consider applying the Ordinary Least Squares estimator to the following regression in the main dataset (HRS), where total health expenditure h is replaced by its imputed counterpart $\hat{h}$:*

$$\hat{h} = w\,\alpha + D\,\beta_2 + u$$

where $\hat{h}$ is imputed from a donor dataset (MEPS) via the regression equation:

$$p = h\,\gamma + D\,\beta_1 + \epsilon$$

and defined as in (21). *Under Assumption B.2, the OLS estimator $\hat{\alpha}$ is consistent.*

Proof. The first step is to show consistency of $\hat{\gamma}$. Define:

$$\hat{\gamma} = \left(\ddot{h}_i'\ddot{h}_i\right)^{-1}\ddot{h}_i'\ddot{p}_i$$

where $i \in \{M, H\}$. From (29), rewrite $\hat{\gamma}$ as:

$$\begin{aligned}
\hat{\gamma} &= \left(\ddot{h}_i'\ddot{h}_i\right)^{-1}\ddot{h}_i'\ddot{h}_i\,\gamma + \left(\ddot{h}_i'\ddot{h}_i\right)^{-1}\ddot{h}_i'\,\epsilon_i \\
&= \gamma + \left(\frac{\ddot{h}_i'\ddot{h}_i}{n_i}\right)^{-1}\frac{\ddot{h}_i'\,\epsilon_i}{n_i} \\
&= \gamma + \left(\frac{\ddot{h}_i'\ddot{h}_i}{n_i}\right)^{-1}\frac{h_i'\,(I - D_i(D_i'D_i)^{-1}D_i')\,\epsilon_i}{n_i}
\end{aligned}$$

By the law of large numbers and Assumption B.2, which implies both $\mathbb{E}[h'\epsilon] = 0$ and $\mathbb{E}[D'\epsilon] = 0$, we have that $\frac{\ddot{h}_i'\ddot{h}_i}{n_i}$ converges to a finite positive definite matrix and $\frac{\ddot{h}_i'\epsilon_i}{n_i} \to 0$, yielding consistency:

$$p\lim\,\hat{\gamma} = \gamma$$

The next step is to show consistency of $\hat{\beta}_1$. By definition:

$$\hat{\beta}_1 = (D_i'D_i)^{-1}D_i'(p_i - h_i\hat{\gamma})$$

Substituting $p_i = h_i\gamma + D_i\beta_1 + \epsilon_i$ from (28):

$$\hat{\beta}_1 = (D_i'D_i)^{-1}D_i'(h_i\gamma + D_i\beta_1 + \epsilon_i - h_i\hat{\gamma})$$

$$= \beta_1 + (D_i'D_i)^{-1}D_i'h_i(\gamma - \hat{\gamma}) + (D_i'D_i)^{-1}D_i'\epsilon_i$$

By the law of large numbers and Assumption B.2, $\frac{D_i'\epsilon_i}{n_i} \to 0$. Combined with $p\text{lim } \hat{\gamma} = \gamma$, the second and third terms vanish, yielding consistency:

$$p\text{lim } \hat{\beta}_1 = \beta_1$$

Next, we characterize the imputed value $\hat{h}_i$ in terms of the true h_i. By definition:

$$\hat{h}_i = \frac{p_i - D_i\hat{\beta}_1}{\hat{\gamma}} = h_i\frac{\gamma}{\hat{\gamma}} + \frac{D_i(\beta_1 - \hat{\beta}_1)}{\hat{\gamma}} + \frac{\epsilon_i}{\hat{\gamma}}$$

Applying the residual maker:

$$\ddot{\hat{h}}_i = \ddot{h}_i\frac{\gamma}{\hat{\gamma}} + \frac{\ddot{\epsilon}_i}{\hat{\gamma}}$$

Finally, from (29) we define the OLS estimator of α as:

$$\hat{\alpha} = \left(\ddot{w}_i'\ddot{w}_i\right)^{-1}\ddot{w}_i'\ddot{\hat{h}}_i$$

Substituting $\ddot{\hat{h}}_i = \ddot{h}_i\frac{\gamma}{\hat{\gamma}} + \frac{\ddot{\epsilon}_i}{\hat{\gamma}}$:

$$\hat{\alpha} = \frac{\gamma}{\hat{\gamma}}\left(\ddot{w}_i'\ddot{w}_i\right)^{-1}\ddot{w}_i'\ddot{h}_i + \frac{1}{\hat{\gamma}}\left(\ddot{w}_i'\ddot{w}_i\right)^{-1}\ddot{w}_i'\ddot{\epsilon}_i$$

Substituting $\ddot{h}_i = \ddot{w}_i\alpha + \ddot{u}_i$:

$$\hat{\alpha} = \frac{\gamma}{\hat{\gamma}}\alpha + \frac{\gamma}{\hat{\gamma}}\left(\ddot{w}_i'\ddot{w}_i\right)^{-1}\ddot{w}_i'\ddot{u}_i + \frac{1}{\hat{\gamma}}\left(\ddot{w}_i'\ddot{w}_i\right)^{-1}\ddot{w}_i'\ddot{\epsilon}_i$$

Expanding using the residual maker:

$$\frac{\ddot{w}_i'\ddot{u}_i}{n_i} = \frac{w_i'\left(I - D_i(D_i'D_i)^{-1}D_i'\right)u_i}{n_i} = \frac{w_i'u_i}{n_i} - \frac{w_i'D_i(D_i'D_i)^{-1}D_i'u_i}{n_i}$$

$$\frac{\ddot{w}_i'\ddot{\epsilon}_i}{n_i} = \frac{w_i'\left(I - D_i(D_i'D_i)^{-1}D_i'\right)\epsilon_i}{n_i} = \frac{w_i'\epsilon_i}{n_i} - \frac{w_i'D_i(D_i'D_i)^{-1}D_i'\epsilon_i}{n_i}$$

By the law of large numbers, both expressions converge to zero under Assumption B.2, which implies $\mathbb{E}[w'u] = 0$, $\mathbb{E}[D'u] = 0$, $\mathbb{E}[D'\epsilon] = 0$, and $\mathbb{E}[w'\epsilon] = 0$.[44] Combined with $p\lim\ \hat{\gamma} = \gamma$, yielding:

$$p\lim\ \hat{\alpha} = \alpha$$

□

B.3 Data and Sample Selection

Having established the consistency of the imputation estimator, we now describe the two datasets used to implement it empirically. The Medical Expenditure Panel Survey (MEPS) serves as the donor dataset from which we estimate the relationship between total and out-of-pocket health spending, while the Health and Retirement Study (HRS) is the main dataset in which we apply the imputation–We already discussed the properties of HRS in Appendix A.

Since 1996, the Medical Expenditure Panel Survey has conducted large-scale surveys of individuals, families, and their healthcare providers, including doctors, hospitals, and pharmacies. MEPS collects data on the specific health services that Americans use, how frequently they use them, how much they cost, and how they are paid for. It currently consists of two major components: the Household Component and the Insurance Component. Data from households and their members in the Household Component are complemented by data from their medical providers, distinguishing MEPS from representative surveys that rely solely on self-reported expenditures. As discussed in

[44]Since $h = w\alpha + D\beta_2 + u$, the exogeneity assumption $\mathbb{E}[h'\epsilon] = 0$ implies $\alpha\mathbb{E}[w'\epsilon] + \beta_2\mathbb{E}[D'\epsilon] + \mathbb{E}[u'\epsilon] = 0$, which together with $\mathbb{E}[D'\epsilon] = 0$ and $\mathbb{E}[u'\epsilon] = 0$ yields $\mathbb{E}[w'\epsilon] = 0$.

the previous section, this provider-verified structure alleviates concerns about measurement error bias, which is the key property that motivates our use of MEPS as the donor dataset.

In this study, we use the Household Component of MEPS. A nationally representative subsample of households that participated in the National Health Interview Survey is used to collect data from families and individuals in selected communities across the United States. MEPS collects detailed information about each household member, including demographic characteristics such as sex, race, marital status, and education, as well as information on both total and out-of-pocket health spending. Through an overlapping panel design, data are collected for two calendar years for each panel of sample households, with a total of five rounds of interviews conducted over a two-and-a-half-year period. To be consistent with HRS sampling, we use MEPS Longitudinal Data Files, which follow respondents across the two years of each panel.

Sample Selection

The first major distinction between the two surveys is that HRS records long-term health expenditures. To mitigate this discrepancy, we restrict the sample to individuals who have resided in nursing homes for one night or fewer, removing 20,629 observations (15.5 percent of the initial sample). Second, since out-of-pocket spending in HRS is imputed by RAND, there are notable outliers that may bias results. Following the RAND HRS documentation, we exclude observations with more than $20,000 in out-of-pocket health spending, removing 2,327 observations (1.7 percent). Third, since all individuals above age 85 are top-coded as 85 in MEPS, we restrict the sample to individuals aged between 51 and 84, excluding 7,144 observations (5.4 percent). Fourth, since we apply a log transformation to health spending variables, individuals with zero total or out-of-pocket spending are excluded, removing 12,695 observations (9.5 percent). These zeros are disproportionately concentrated in HRS, consistent with differences in how spending is recorded across the two surveys. Finally, due to differences in race definitions and oversampling practices across the two datasets, we focus exclusively on individuals identified as Black or White, dropping 6,865 observations (5.1 percent). After all restrictions, the analysis sample contains 83,693 observations. The analysis covers the period 2006–2016, corresponding to the waves for which both datasets are jointly available and comparable.

Table A7 compares demographic means across the two datasets for the period 2006–2016. Given

TABLE A7. Comparing Demographic Means

HRS

	2006	2008	2010	2012	2014	2016
Sex	0.45	0.46	0.46	0.46	0.46	0.46
Race	0.92	0.91	0.91	0.91	0.92	0.91
Married	0.72	0.71	0.71	0.70	0.70	0.71
University Degree	0.28	0.29	0.32	0.33	0.34	0.35
Highschool Degree	0.87	0.88	0.90	0.91	0.92	0.93
Age	63.84	65.06	62.82	63.92	65.24	63.88

MEPS

	2006	2008	2010	2012	2014	2016
Sex	0.46	0.47	0.47	0.47	0.47	0.47
Race	0.91	0.90	0.90	0.89	0.89	0.89
Married	0.66	0.64	0.64	0.63	0.61	0.64
University Degree	0.34	0.36	0.39	0.40	0.42	0.41
Highschool Degree	0.86	0.86	0.88	0.88	0.85	0.89
Age	63.46	63.49	63.32	63.43	63.62	64.20

This table compares main regressors in the imputation process. Sex takes the value 1 if the respondent is male, and zero otherwise. Race takes the value of 1 if the respondent is white. Married represents the share of married couples and the last two variables illustrate the share of respondents with a university and high school degree.

that HRS waves and longitudinal MEPS panels each span two years, we analyze the data in corresponding two-year periods.

The gender and racial compositions of the two samples are closely aligned throughout the period. Sex is virtually identical across all waves, and the share of White respondents follows a similar pattern in both datasets, ranging between 89 and 92 percent.

Differences are more pronounced for marital status and education. HRS respondents are approximately 6–9 percentage points more likely to be married than MEPS respondents, and MEPS respondents are 5–8 percentage points more likely to hold a university degree. Average age is comparable across the two datasets, though HRS exhibits more variation across waves—ranging from approximately 63 to 65, while MEPS remains more stable around 63–64. To formally assess whether these differences are statistically significant, we estimate the following regression separately for each wave:

$$D_i = \alpha + \beta \, Meps_i + \epsilon_i$$

where D_i represents each demographic variable and $Meps_i$ is an indicator for belonging to the MEPS dataset. Results are reported in Table A8.

As expected, differences in sex are statistically insignificant across all waves. Differences in race

Table A8. Differences in MEPS and HRS Demographics

	2006	2008	2010	2012	2014	2016
Male	0.01	0.01	0.00	0.01	0.01	0.01
	(0.31)	(0.43)	(0.68)	(0.30)	(0.24)	(0.55)
White	-0.01	-0.01	-0.01	-0.01**	-0.02***	-0.02***
	(0.18)	(0.07)	(0.07)	(0.01)	(0.00)	(0.00)
Married	-0.06***	-0.07***	-0.07***	-0.07***	-0.09***	-0.07***
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
University Degree	0.06***	0.07***	0.07***	0.07***	0.08***	0.06***
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Highschool Degree	-0.02**	-0.02***	-0.02***	-0.03***	-0.07***	-0.05***
	(0.02)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Age	-0.37*	-1.57***	0.51***	-0.50**	-1.62***	0.32
	(0.07)	(0.00)	(0.01)	(0.01)	(0.00)	(0.11)
OOP	-94.35	-108.80*	-523.55***	-603.19***	-503.77***	-601.95***
	(0.11)	(0.08)	(0.00)	(0.00)	(0.00)	(0.00)

This table reports the estimated $\hat{\beta}$ from regressing each demographic variable on a MEPS indicator, estimated separately for each wave. Numbers in parentheses are p-values. Stars follow the convention: $^{*}\, p < 0.10$, $^{**}\, p < 0.05$, $^{***}\, p < 0.01$.

are small in magnitude (1–2 percentage points), though statistically significant in some later waves. Differences in marital status and education are statistically significant, and age differences reflect the cohort structure of the HRS and vary in sign across waves. In addition, differences in out-of-pocket spending are statistically significant and grow over time. Overall, these patterns point to the necessity of the correction proposed in equation (25).

Imputation Consistency Within MEPS (donor)

Table A9 compares the true and imputed total expenditure in log terms.[45]. The imputed values successfully reproduce the mean values, with differences of less than 0.1 in all years. This aligns with the theoretical prediction that, with consistent estimators, the imputation should closely

[45] A linear specification ignores the concavity of the relationship between out-of-pocket and total spending: the out-of-pocket share of total spending declines with the level of expenditure, reflecting deductibles, coinsurance, and out-of-pocket maximums. The log transformation linearizes this relationship.

replicate the mean.[46]

TABLE A9. True vs. Imputed Total Expenditure — MEPS

	Data				Imputation			
	mean	sd	iqr	skewness	mean	sd	iqr	skewness
2006	8.6	1.4	1.6	-0.10	8.6	1.7	2.0	-0.19
2008	8.7	1.4	1.7	-0.13	8.8	1.8	2.1	-0.20
2010	8.7	1.5	1.9	-0.11	8.7	2.0	2.4	-0.16
2012	8.6	1.6	1.9	-0.12	8.7	2.0	2.5	-0.19
2014	8.7	1.5	1.9	-0.13	8.8	2.1	2.5	-0.22
2016	8.8	1.6	1.9	-0.11	8.9	2.3	2.8	-0.17

This table summarizes different distribution moments in actual and imputed total medical expenditure. The left box represents actual moments and the right box represents the imputed data moments. In each box, the first column represents mean, the second represents standard deviation, the third represents interquantile range and the forth represents quantile based skewness.

In terms of standard deviation, the differences are more pronounced, and our imputation does not consistently reproduce the observed standard deviation. This outcome is consistent with equation (23), where we noted that even when estimates are consistent, the predicted standard deviation may differ due to the residual variance term $\sigma_\epsilon^2/\gamma^2$. Notably, the gap between the true and imputed standard deviations widens over time—from approximately 0.3 in 2006 to 0.7 in 2016—suggesting that the residual variance may be increasing across waves.

The widening gap between true and imputed standard deviations coincides with the entry of new cohorts into the HRS sample. The HRS employs a steady-state design, refreshing the sample every six years with younger birth cohorts aged 51–56 at enrollment. In 2010, the Mid Baby Boomers (born 1954–1959) entered the sample, and in 2016, the Late Baby Boomers (born 1960–1965) were added. These refreshment waves introduce respondents with systematically different health

[46]Including an unrestricted constant in the log specification predicts that total health spending is less than out-of-pocket spending when the latter measure is relatively small. To avoid this issue, the demographic controls subsume the role of the intercept, allowing the level of out-of-pocket spending to vary across individuals through observed characteristics rather than an undifferentiated constant.

TABLE A10. Imputed Total Expenditure - MEPS vs. HRS

	MEPS				HRS				Mean Difference		SD Difference		Adjusted HRS
	mean	sd	iqr	skewness	mean	sd	iqr	skewness	Out of Pocket	Total	Out of Pocket	Total	mean
2006	8.64	1.72	2.00	-0.19	8.55	1.83	2.26	-0.16	-0.06	-0.09	-0.06	-0.09	8.63
2008	8.75	1.85	2.09	-0.20	8.79	1.79	2.24	-0.14	0.04	0.03	0.04	0.03	8.73
2010	8.74	2.02	2.44	-0.16	9.11	1.91	2.47	-0.10	0.27	0.36	0.27	0.36	8.70
2012	8.66	2.04	2.49	-0.19	9.10	1.93	2.47	-0.13	0.32	0.44	0.32	0.44	8.61
2014	8.78	2.11	2.55	-0.22	9.14	1.99	2.62	-0.10	0.29	0.35	0.29	0.35	8.70
2016	8.86	2.35	2.83	-0.17	9.34	2.20	2.93	-0.10	0.34	0.48	0.34	0.48	8.78

This table compares imputed total medical expenditure in MEPS and HRS. The first eight columns report the mean, standard deviation, interquartile range, and quantile-based skewness for each dataset. The next four columns decompose the mean and standard deviation differences into contributions from out-of-pocket spending. The final column reports the adjusted HRS mean following Theorem B.2.

profiles and spending patterns than the incumbent sample.

Both interquartile ranges and standard deviations are smaller in the original data than in the imputed data, yet their ratio remains stable across all years and similar across the true and imputed distributions. In the true data, this ratio ranges from approximately 1.14 to 1.19, while in the imputed data it ranges from approximately 1.18 to 1.22, compared to 1.35 for a standard normal distribution. This indicates that our imputation method tends to preserve the relative spread of the actual data, with dispersion somewhat more concentrated than a normal distribution. Skewness levels are also comparable, although we rely on quantile-based skewness to mitigate the influence of outliers. The imputed data exhibits a slightly greater left skew in every wave.

Imputing Total Medical Spending in HRS

Table A10 compares the imputed total medical expenditure in MEPS and HRS. The mean values differ across the two datasets, with the gap widening over time. In 2006 and 2008, the difference in average imputed total spending is less than 10 percent. However, beginning in 2010, this gap increases substantially, reaching 30 to 48 percent by 2016.

Despite the differences in means, the dispersion of imputed spending is similar across the two datasets. The difference in standard deviation is less than 0.1 in all years, and the interquartile ranges follow a comparable pattern. This suggests that the imputed total spending in HRS preserves the spread observed in MEPS, consistent with the theoretical results in Section B.1. The imputed

HRS distribution exhibits slightly less left skewness than MEPS, though both distributions remain moderately left-skewed throughout the sample period.

Consistent with Equation (25), differences in out-of-pocket spending account for the bulk of the divergence in imputed means. As shown in Table A10, approximately 80 percent of the gap in mean imputed total spending is attributable to differences in out-of-pocket expenditures across the two samples. A similar pattern holds for the standard deviation.

The final column of Table A10 reports the adjusted HRS mean, computed following Theorem B.2. After applying the out-of-pocket adjustment, the corrected HRS means closely track the imputed MEPS values across all waves. Figure 11 illustrates this alignment visually, confirming that the adjustment successfully removes the systematic bias induced by differences in out-of-pocket spending distributions.

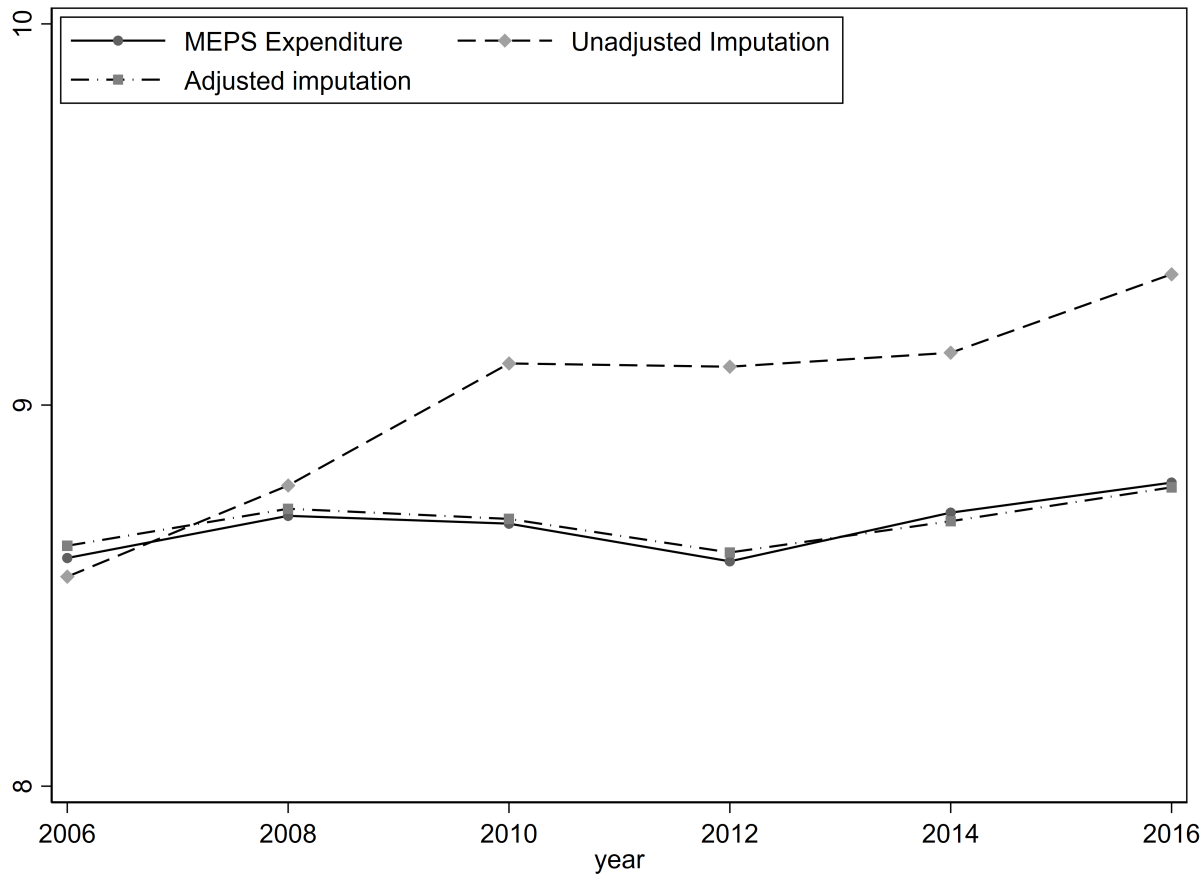
B.4 Medical Spending–Wealth Gradient

Having established the validity of our imputation procedure, we now examine the relationship between total medical spending and wealth. A key advantage of imputing total medical expenditure in the HRS is access to detailed wealth measures unavailable in MEPS. While MEPS provides high-quality medical spending data verified by healthcare providers, it contains limited information on household assets. The HRS, by contrast, collects comprehensive data on wealth holdings, including housing, financial assets, pensions, and business equity. By combining imputed medical spending with the rich wealth data in the HRS, we can characterize the spending-wealth gradient across the wealth distribution.

Sample Construction

We analyze the relationship between medical spending and wealth separately for single-person and couple households. This separation is necessary because wealth in the HRS is measured at the household level, while medical spending is recorded for each individual. For single-person households, the individual and household unit coincide, so no aggregation is required. For couples, we must aggregate individual medical expenditures to the household level to match the unit of observation for wealth.

Figure 11. Total Medical Spending Mean Comparison



Notes: This figure plots the mean of imputed total medical expenditure (in logs) across waves for three series: imputed MEPS, imputed HRS, and adjusted HRS. The adjusted HRS series applies the correction from Theorem B.2, subtracting the scaled difference in mean out-of-pocket spending between the two datasets. After adjustment, the HRS means closely align with the MEPS values, confirming that differences in out-of-pocket spending distributions account for the divergence in unadjusted means.

For couple households, we aggregate imputed medical spending as follows. Let $\hat{h}_1$ and $\hat{h}_2$ denote the imputed log medical expenditure for each spouse. Since log values cannot be summed directly, we first convert to levels, sum across spouses, and then take the logarithm:

$$\hat{h}_{hh} = \log\Big(\exp(\hat{h}_1) + \exp(\hat{h}_2)\Big) \tag{30}$$

This household-level measure, $\hat{h}_{hh}$, represents total medical spending for the couple and is directly comparable to household wealth.

We impose several sample restrictions. First, we limit the analysis to waves 8–13 (2006–2016), corresponding to the period for which imputed medical expenditure is available. Second, we retain only observations satisfying the imputation sample criteria: individuals aged 51–84, out-of-pocket spending at or below $20,000, positive total and out-of-pocket spending, and racial classification as Black or White. Third, we exclude individuals residing in nursing homes for more than one night. These restrictions ensure comparability between the HRS analysis sample and the MEPS donor sample used for imputation.

Additionally, we impose restrictions specific to each household type. For couples, we require that both spouses respond in each wave, that neither spouse remarries during the panel, and that the household does not split due to divorce or separation. These restrictions ensure that we observe the same couple unit throughout the panel. For singles, we exclude individuals who marry during the observation period.

Summary Statistics

Table A11 presents summary statistics for the analysis sample. The couple sample comprises 24,315 household-wave observations spanning 7,468 unique households, while the single sample contains 10,869 individual-wave observations from 4,379 unique individuals. All monetary values are expressed in real terms. For couples, demographic variables are measured at the household level: age is the average of both spouses, while education and race indicators capture whether either or both spouses possess the characteristic.

The comparison of out-of-pocket and total medical expenditure across household types is informative about the imputation procedure. Couples exhibit higher log medical spending than singles

Table A11. Summary Statistics

	Couples		Singles	
	Mean	SD	Mean	SD
Medical Spending				
Log total medical expenditure	9.63	1.97	9.23	2.21
Log Out-of-pocket expenditure	8.12	1.17	6.98	1.36
Wealth				
Total wealth ($000s)	630.76	1,342.09	282.51	2,907.87
Demographics				
Age (average of spouses / individual)	65.21	9.16	64.30	9.23
Male (%)	—	—	26	44
Either spouse college (%) / College (%)	47	50	32	47
Both spouses college (%)	21	41	—	—
Either spouse White (%) / White (%)	87	34	65	48
Both spouses White (%)	81	39	—	—
Household-wave observations	24,315		10,869	
Unique households/individuals	7,468		4,379	

Notes: Sample restricted to waves 8–13 (2006–2016) with non-missing imputed medical expenditure. Wealth values in thousands of real dollars. For couples, the unit of observation is the household-wave; medical expenditure is aggregated across both spouses, and demographic variables are measured at the household level.

on both margins: log total medical expenditure averages 9.63 for couples versus 9.23 for singles, a gap of roughly 0.3 log points, while log out-of-pocket expenditure averages 8.12 for couples versus 6.98 for singles, a gap of 1.15 log points.

Total wealth is also different between these two groups. Couples' net worth averages to $630.76 thousand, while singles hold $282.51 on average. As discussed in Appendix A, couples hold more diversified and higher total wealth compared to singles, which could explain the reversal pattern in total health spending.

The demographic composition of the two samples also differs. Singles are disproportionately female (74%) and less likely to be White (65% versus 81–87% for couples). Educational attainment differs somewhat: 47% of couple households have at least one college-educated spouse, though only 21% have both spouses with college degrees, compared to 32% of singles holding a degree. These compositional differences motivate our decision to analyze the two samples separately rather than pooling them with a household-type indicator.

Estimation

To characterize the medical spending–wealth gradient, we estimate the following pooled specification separately for couple and single households:

$$\hat{h}_{it} = \alpha + \sum_{d=2}^{10} \theta_d \cdot \mathbf{1}[W_{it} \in \text{decile } d] + D_{it}\beta + \delta_t + u_{it} \tag{31}$$

where $\hat{h}_{it}$ denotes imputed log total medical expenditure for household i in wave t, W_{it} is household net worth, D_{it} collects demographic controls, and δ_t are wave fixed effects. The coefficients $\{\theta_d\}_{d=2}^{10}$ trace out the spending gradient across the wealth distribution, with the first (poorest) decile as the reference group. Standard errors are clustered at the household level to account for repeated observations on the same unit across waves.

Wealth deciles are constructed by pooling all household-wave observations within each sample and sorting by net worth. This approach maximizes statistical power relative to constructing deciles separately within each wave. Since all monetary values are CPI-adjusted, nominal price-level differences across waves do not affect the ranking. The wave fixed effects δ_t absorb any remaining aggregate shifts in medical spending—including those arising from policy changes such as the Affordable Care Act and compositional shifts from HRS refreshment cohorts—ensuring that the decile coefficients reflect cross-sectional variation in wealth rather than time-series trends.

For couples, the demographic controls include the average age of both spouses, an indicator for whether either spouse holds a college degree, and an indicator for whether either spouse is White. For singles, the controls are individual age, college degree, race, and gender. These variables correspond to the regressors used in the imputation equation estimated in Section B.1.

Results

Table A12 reports the mean wealth, the estimated percentage difference in total medical spending, and the corresponding difference in out-of-pocket spending for each wealth decile, separately for couples and singles. By comparing the two spending measures estimated under the same specification, we can assess the extent to which out-of-pocket expenditure understates the medical spending–wealth gradient.

For singles, both measures exhibit a U-shaped pattern at the bottom of the wealth distribution, with deciles 2 and 3 spending less than the poorest decile. The two gradients track each other more closely than in the couple sample: at the tenth decile, total spending is 37 percent above the reference group compared to 35 percent for out-of-pocket, a gap of only 3 percentage points. Figure 12.a illustrates this pattern. The narrower gap for singles is consistent with less generous insurance coverage among single-person households, which would cause out-of-pocket spending to more closely approximate total expenditure.

For couples, the total spending gradient is rising from 27 percent at the second decile to 82 percent at the tenth decile relative to the poorest households. The out-of-pocket gradient is substantially flatter: it ranges from 15 to 27 percent across deciles 2 through 9 and rises to 47 percent only at the top decile. Figure 12.b displays both gradients. The gap between the two measures widens steadily across the wealth distribution, from approximately 12 percentage points at the second decile to 35 percentage points at the tenth.

Two findings emerge from 12. First, out-of-pocket spending understates the medical spending–wealth gradient for couples: at the top decile, the total gradient (82.5%) is nearly twice as steep as the out-of-pocket gradient (47.4%), a gap of roughly 35 percentage points. Studies relying solely

Table A12. Medical Spending by Wealth Decile: Total vs. Out-of-Pocket

	Couples			Singles		
Decile	Mean Wealth ($000s)	Total (%)	OOP (%)	Mean Wealth ($000s)	Total (%)	OOP (%)
1	−11.2	ref.	ref.	−26.5	ref.	ref.
2	40.9	27.1%	15.2%	0.1	−47.3%	−39.6%
3	91.5	27.1%	19.3%	4.0	−24.5%	−16.8%
4	151.6	32.2%	16.4%	18.2	1.3%	0.7%
5	226.8	27.0%	18.1%	46.2	5.1%	5.3%
6	327.3	45.2%	23.4%	85.2	27.0%	21.6%
7	472.1	45.3%	20.3%	141.6	12.5%	14.6%
8	681.5	42.6%	21.2%	237.7	7.6%	12.1%
9	1,084.9	58.1%	27.4%	432.9	27.4%	27.0%
10	3,243.0	82.5%	47.4%	1,887.3	37.4%	34.8%

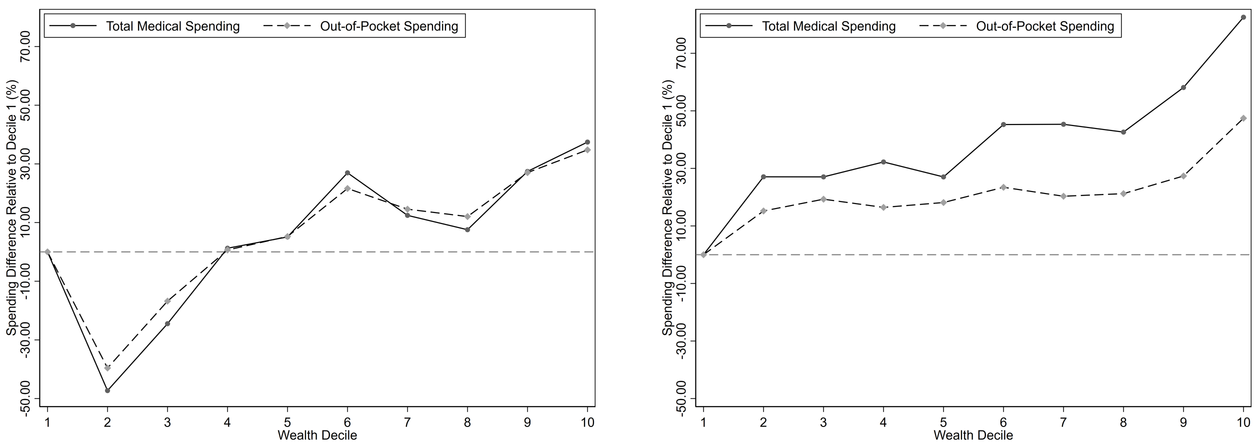
Notes: Mean Wealth reports the average net worth within each decile in thousands of real dollars. Total and OOP columns report the percentage difference in spending relative to decile 1. Total refers to imputed total medical expenditure; OOP refers to observed out-of-pocket medical expenditure. Both regressions include wave fixed effects and demographic controls (age, education, race for both; gender for singles). Standard errors clustered at the household level.

on out-of-pocket data would conclude that medical spending is only modestly related to wealth for this group, when in fact the total cost gradient is substantially steeper.

Second, the pattern differs sharply for singles. The total and out-of-pocket gradients track closely across the distribution, with a gap of only 2.6 percentage points at the top decile. Empirically, it provides direct support for the exclusion restriction underlying the imputation: if wealth affected out-of-pocket spending through the insurance share (conditional on total spending and demographics), we would expect to see a wedge between the two gradients. Among singles, this wedge is small, suggesting the residual correlation between wealth and the imputation error is limited in that sample.

Figure 12. Total vs. Out-of-Pocket Spending by Wealth Decile

a. Singles b. Couples



Notes: Panel a. plots the estimated percentage difference in spending relative to decile 1 for single-person households, using both imputed total medical expenditure (solid) and observed out-of-pocket expenditure (dashed). Both regressions include wave fixed effects and demographic controls, with standard errors clustered at the individual level. Panel b. applies the same method to couple households.

C Theoretical Appendix

C.1 Monotonicity of the Value Function and Policy Functions in (BL-L)

This appendix proves the monotonicity claim invoked in the proof of Theorem 3.2: the low-state value function $\underline{V}(b)$ and its policy functions $\underline{c}(b)$, $\underline{m}(b)$, and $\underline{b}'(b)$ are all non-decreasing in wealth b. Throughout, $\underline{V}$ is maintained to be concave, as assumed in the main text. Let $\underline{V}^n$ denote the n-th iterate of the Bellman operator associated with (BL-L), with $\underline{V}^0(b) = u\left(\underline{c}_0(b)\right)$ the terminal (static) problem in which all resources are consumed, and

$$\underline{V}^{n+1}(b) = \max_{c,m,b'} \left\{u(c) + \rho\, \chi(m)\, \underline{V}^n(b')\right\} \quad \text{s.t.} \quad c + m + b' = (1+r)b + y, \;\; b' \geq \underline{b}.$$

We use throughout the following first-order and envelope conditions of this problem, letting $\underline{c}(b)$, $\underline{m}(b)$, and $\underline{b}'(b)$ denote the policy functions solving the $n+1$ problem at wealth b:

$$\underline{V}_b^{n+1}(b) = u_c\left(\underline{c}(b)\right), \tag{32}$$

$$\underline{V}_b^{n+1}(b) = \rho\,\chi\left(\underline{m}(b)\right)\,\underline{V}_b^{n}\left(\underline{b}'(b)\right), \tag{33}$$

$$\rho\,\chi_m\left(\underline{m}(b)\right)\,\underline{V}^{n}\left(\underline{b}'(b)\right) = u_c\left(\underline{c}(b)\right). \tag{34}$$

Equations (32) and (33) both equate to the multiplier on the budget constraint—the envelope condition with respect to b, and the first-order condition with respect to b', respectively—while (34) is the first-order condition with respect to m.

Lemma C.1 *$\underline{c}(b)$ is non-decreasing in b.*

Proof. Since $\underline{V}^{n+1}$ is concave, $\underline{V}_b^{n+1}$ is non-increasing in b. By (32), this means $u_c\left(\underline{c}(b)\right)$ is non-increasing in b; since u_c is strictly decreasing in consumption, $\underline{c}(b)$ must be non-decreasing in b. □

Lemma C.2 *Suppose $\underline{V}^n$ is concave and non-decreasing. Then $\underline{m}(b)$ and $\underline{b}'(b)$ are non-decreasing in b.*

Proof. Take $b_1 > b_2$. Since $\underline{V}^{n+1}$ is concave, $\underline{V}_b^{n+1}(b_1) < \underline{V}_b^{n+1}(b_2)$; substituting (33) at b_1 and b_2 gives

$$\rho\,\chi\left(\underline{m}(b_1)\right)\,\underline{V}_b^{n}\left(\underline{b}'(b_1)\right) < \rho\,\chi\left(\underline{m}(b_2)\right)\,\underline{V}_b^{n}\left(\underline{b}'(b_2)\right). \tag{$*$}$$

Case 1: $\underline{m}(b_1) \geq \underline{m}(b_2)$. Since χ is increasing in m, $\chi\left(\underline{m}(b_1)\right) \geq \chi\left(\underline{m}(b_2)\right) > 0$. For ($*$) to hold, this forces $\underline{V}_b^{n}\left(\underline{b}'(b_1)\right) < \underline{V}_b^{n}\left(\underline{b}'(b_2)\right)$. Since $\underline{V}^n$ is concave, $\underline{V}_b^n$ is decreasing, so this implies $\underline{b}'(b_1) > \underline{b}'(b_2)$: both policy functions are increasing in this case.

Case 2: $\underline{m}(b_1) < \underline{m}(b_2)$. By Lemma C.1, $\underline{c}(b_1) \geq \underline{c}(b_2)$, so $u_c\left(\underline{c}(b_1)\right) \leq u_c\left(\underline{c}(b_2)\right)$. By (34),

$$\chi_m\left(\underline{m}(b_1)\right)\,\underline{V}^{n}\left(\underline{b}'(b_1)\right) \leq \chi_m\left(\underline{m}(b_2)\right)\,\underline{V}^{n}\left(\underline{b}'(b_2)\right).$$

Since χ_m is decreasing in m (the health production function exhibits diminishing returns), $\underline{m}(b_1) < \underline{m}(b_2)$ implies $\chi_m\left(\underline{m}(b_1)\right) > \chi_m\left(\underline{m}(b_2)\right)$, which forces $\underline{V}^{n}\left(\underline{b}'(b_1)\right) < \underline{V}^{n}\left(\underline{b}'(b_2)\right)$; since $\underline{V}^n$ is non-decreasing, this in turn implies $\underline{b}'(b_1) < \underline{b}'(b_2)$. Therefore, $\underline{m}(b)$ and $\underline{b}'(b)$ would both be decreasing in b; since $\underline{m}(b) \geq 0$ and $\underline{b}'(b) \geq \underline{b}$ are bounded below, they would converge to their respective floors, $\underline{m}(b) \to 0$ and $\underline{b}'(b) \to \underline{b}$, as $b \to \infty$.

At the same time, since $\underline{c}(b)$, $\underline{m}(b)$, and $\underline{b}'(b)$ must exhaust the budget constraint $\underline{c}(b) + \underline{m}(b) + \underline{b}'(b) = (1+r)b + y$, and the right-hand side diverges while $\underline{m}(b)$ and $\underline{b}'(b)$ converge to finite constants, $\underline{c}(b) \to \infty$. Evaluating (34) in this limit, the right-hand side $u_c\left(\underline{c}(b)\right) \to 0$, while the left-hand side converges to $\rho\, \chi_m(0)\, \underline{V}^n(\underline{b}) > 0$: a strictly positive constant, since $\chi_m(0)$ is finite and strictly positive by (1), and $\underline{V}^n(\underline{b})$ is some positive constant. This contradicts the requirement that (34) hold with equality for every b. Case 2 is therefore impossible, and $\underline{m}(b)$, $\underline{b}'(b)$ must be non-decreasing in b. $\square$

Proposition C.3 *$\underline{V}(b)$, $\underline{c}(b)$, $\underline{m}(b)$, and $\underline{b}'(b)$ are all non-decreasing in b.*

Proof. By induction on n. The base case $\underline{V}^0(b) = u\left(\underline{c}_0(b)\right)$ is non-decreasing since all resources are consumed in the terminal problem. Suppose $\underline{V}^n$ is concave and non-decreasing. By Lemmas C.1 and C.2, $\underline{c}(b)$, $\underline{m}(b)$, and $\underline{b}'(b)$ are non-decreasing in b; since χ is increasing and $\underline{V}^n$ is non-decreasing, the composition $\chi\left(\underline{m}(b)\right) \underline{V}^n\left(\underline{b}'(b)\right)$ is non-decreasing in b, and therefore so is

$$\underline{V}^{n+1}(b) = u\left(\underline{c}(b)\right) + \rho\, \chi\left(\underline{m}(b)\right)\, \underline{V}^n\left(\underline{b}'(b)\right).$$

Induction, together with the convergence of $\underline{V}^n$ to $\underline{V}$ established in the proof of Theorem 3.2, implies that the infinite-horizon value function $\underline{V}$, and its associated policy functions $\underline{c}$, $\underline{m}$, and $\underline{b}'$, are all non-decreasing in wealth. $\square$

C.2 BAH: Decreasing Saving Function

This appendix shows that in the BAH framework, households should start to dissave after some wealth threshold. This is equivalent to the Huggett (1993) appendix which follow Schechtman (1976).

Lemma C.4 *For all $b > \underline{b}$, $\underline{b}'(b) \leq b$.*

Proof. The Euler equation for the low-type with wealth b is

$$u'\left(\underline{c}\left(b\right)\right)=\frac{\rho}{\left(1+r\right)}\left[\underline{\pi}u'\left(\underline{c}\left(\underline{b}'\left(b\right)\right)\right)+\left(1-\underline{\pi}\right)u'\left(\bar{c}\left(\underline{b}'\left(b\right)\right)\right)\right]$$
$$<\underline{\pi}u'\left(\underline{c}\left(\underline{b}'\left(b\right)\right)\right)+\left(1-\bar{\pi}\right)u'\left(\bar{c}\left(\underline{b}'\left(b\right)\right)\right). \quad (35)$$

Since $\underline{c}\left(b\right)\leq\bar{c}\left(b\right)$, this last inequality holds only if

$$u'\left(\underline{c}\left(\underline{b}'\left(b\right)\right)\right)>u'\left(\underline{c}\left(b\right)\right),$$

or $\underline{c}\left(\underline{b}'\left(b\right)\right)<\underline{c}\left(b\right)$. Since $\underline{c}\left(b\right)$ is non-decreasing in b, this is true only if $\underline{b}'\left(b\right)<b$. □

Note that $\underline{b}'\left(-\underline{b}\right)\leq-\underline{b}$ can only hold as a strict equality. As a result, $\underline{b}'\left(b\right)$ must pass through the point $\left(-\underline{b},-\underline{b}\right)$ and lie under the 45-degree line for all values of b.

Lemma C.5 *When $\bar{b}'\left(b\right)>b$, then $\bar{c}\left(b\right)-\underline{c}\left(b\right)<\bar{y}-\underline{y}$.*

Proof. By Lemma C.4, we know that $rb+\underline{y}-\underline{c}\left(b\right)\leq0$, or

$$\underline{c}\left(b\right)-\underline{y}\geq rb.$$

On the other hand, for $\bar{b}'\left(b\right)>b$, we need to have $\bar{c}\left(b\right)-\bar{y}<rb$, or

$$-\left[\bar{c}\left(b\right)-\bar{y}\right]>-rb.$$

If we add these two inequalities, we get the intended result. □

A corollary of this is that

$$0<\bar{b}'\left(b\right)-\underline{b}'\left(b\right)=\bar{y}-\underline{y}-\left[\bar{c}\left(b\right)-\underline{c}\left(b\right)\right]<\bar{y}-\underline{y}. \quad (36)$$

Therefore,

$$\underline{b}'\left(b\right)\geq b-\left(\bar{y}-\underline{y}\right). \quad (37)$$

If we let $\kappa:=\left(\bar{y}-\underline{y}\right)$, from Lemma C.5, we can write $\bar{c}\left(b\right)-\kappa<\underline{c}\left(b\right)\leq\bar{c}\left(b\right)$, where the second inequality follows from the monotonicity of optimal consumption. Therefore,

$$u'\left(\underline{c}\left(b\right)\right) < u'\left(\bar{c}\left(b\right) - \kappa\right) \approx u'\left(\bar{c}\left(b\right)\right) - \kappa u''\left(\bar{c}\left(b\right)\right). \tag{38}$$

Let us plug this into the Euler equation for the high-types:

$$\begin{aligned} u'\left(\bar{c}\left(b\right)\right) &= \frac{\rho}{(1+r)}\left[\bar{\pi}u'\left(\bar{c}\left(\bar{b}'\left(b\right)\right)\right) + (1-\bar{\pi})\, u'\left(\underline{c}\left(\bar{b}'\left(b\right)\right)\right)\right] \\ &< \bar{\pi}u'\left(\bar{c}\left(\bar{b}'\left(b\right)\right)\right) + (1-\bar{\pi})\, u'\left(\underline{c}\left(\bar{b}'\left(b\right)\right)\right) \\ &< \bar{\pi}u'\left(\bar{c}\left(\bar{b}'\left(b\right)\right)\right) + (1-\bar{\pi})\left[u'\left(\bar{c}\left(\bar{b}'\left(b\right)\right)\right) - \kappa u''\left(\bar{c}\left(\bar{b}'\left(b\right)\right)\right)\right]. \end{aligned} \tag{39}$$

The last inequality can be rearranged and written as

$$u'\left(\bar{c}\left(b\right)\right) - u'\left(\bar{c}\left(\bar{b}'\left(b\right)\right)\right) < -\left(1-\bar{\pi}\right)\kappa u''\left(\bar{c}\left(\bar{b}'\left(b\right)\right)\right). \tag{40}$$

If we replace for the first and second derivatives of the utility function, assuming a CRRA felicity, we have

$$\frac{1}{\left[\bar{c}\left(b\right)\right]^{\sigma}} - \frac{1}{\left[\bar{c}\left(\bar{b}'\left(b\right)\right)\right]^{\sigma}} < \frac{\left(1-\bar{\pi}\right)\kappa}{\left[\bar{c}\left(\bar{b}'\left(b\right)\right)\right]^{\sigma+1}}. \tag{41}$$

The next theorem uses this inequality to show that the individual's saving rate must eventually become negative, regardless of her level of current income.

Theorem C.6 *There must exist some $\underline{b} < \bar{b} < \infty$ for which, for all $b \geq \bar{b}$, we must have $rb + \bar{y} - \bar{c}\left(b\right) \leq 0$.*

Proof. Multiplying both sides of (41) by $\left[\bar{c}\left(\bar{b}'(b)\right)\right]^{\sigma}$:

$$\frac{\left[\bar{c}\left(\bar{b}'(b)\right)\right]^{\sigma}}{\left[\bar{c}(b)\right]^{\sigma}} - 1 < \frac{(1-\bar{\pi})\kappa}{\bar{c}\left(\bar{b}'(b)\right)} \tag{42}$$

Rearranging:

$$\left[\frac{\bar{c}\left(\bar{b}'(b)\right)}{\bar{c}(b)}\right]^{\sigma} < 1 + \frac{(1-\bar{\pi})\kappa}{\bar{c}\left(\bar{b}'(b)\right)} \tag{43}$$

Taking the limit as $b \to \infty$, and noting that $\bar{c}(b) \to \infty$ and $\bar{c}\left(\bar{b}'(b)\right) \to \infty$, the right-hand side converges to 1, so in the limit:

$$\left[\frac{\bar{c}\left(\bar{b}'(b)\right)}{\bar{c}(b)}\right]^{\sigma} \leq 1 \tag{44}$$

which implies $\bar{c}\left(\bar{b}'(b)\right) \leq \bar{c}(b)$. Since $\bar{c}$ is an increasing function, this implies:

$$\bar{b}'(b) \leq b \tag{45}$$

That is, in the limit for $b \geq \bar{b}$ we have $\bar{b}'(b) - b = rb + \bar{y} - \bar{c}\,(b) \leq 0$ $\square$

References

S. Rao Aiyagari. Uninsured idiosyncratic risk and aggregate saving. *The Quarterly Journal of Economics*, pages 659–684, 1994.

Laurence Ales, Roozbeh Hosseini, and Larry E. Jones. Is there "too much" inequality in health spending across income groups?, 2014.

Laurent Bach, Laurent E Calvet, and Paolo Sodini. From Saving Comes Having? Disentangling the Impact of Saving on Wealth Inequality. 2017.

Jess Benhabib, Alberto Bisin, and Shenghao Zhu. The distribution of wealth and fiscal policy in economies with finitely lived agents. *Econometrica*, 79(1):123–157, 2011.

Jess Benhabib, Alberto Bisin, and Mi Luo. Wealth distribution and social mobility in the us: a quantitative approach. Technical report, National Bureau of Economic Research, 2015.

Truman Bewley. The permanent income hypothesis: a theoretical formulation. *Journal of Economic Theory*, 16(2):252–292, 1977.

Richard W Blundell, Luigi Pistaferri, and Ian Preston. Imputing consumption in the psid using food demand estimates from the cex. Technical report, IFS Working Papers, 2004.

Marco Cagetti. Wealth accumulation over the life cycle and precautionary savings. *Journal of business & economic statistics*, 21(3):339–353, 2003. ISSN 0735-0015.

Marco Cagetti and Mariacristina De Nardi. Entrepreneurship, frictions, and wealth. *Journal of political Economy*, 114(5):835–870, 2006.

Marco Cagetti and Mariacristina De Nardi. Wealth inequality: Data and models. *Macroeconomic Dynamics*, 12(S2):285–313, 2008.

Ana Castaneda, Javier Diaz-Gimenez, and Jose-Victor Rios-Rull. Accounting for the u.s. earnings and wealth inequality. *The Journal of political economy*, 111(4):818–857, 2003. ISSN 0022-3808.

Thomas F. Crossley, Peter Levell, and Stavros Poupakis. Regression with an imputed dependent variable. *Journal of Applied Econometrics*, 37(7):1277–1294, 2022. doi: https://doi.org/10.1002/jae.2921. URL https://onlinelibrary.wiley.com/doi/abs/10.1002/jae.2921.

David M. Cutler and Jonathan Gruber. Does public insurance crowd out private insurance? *Quarterly Journal of Economics*, 111(2):391–430, 1996.

Mariacristina De Nardi. Wealth inequality and intergenerational links. *The Review of Economic Studies*, 71(3):743–768, 2004.

Mariacristina De Nardi and Giulio Fella. Saving and wealth inequality. *Review of Economic Dynamics*, 26:280–300, October 2017. ISSN 10942025. doi: 10.1016/j.red.2017.06.002. URL https://linkinghub.elsevier.com/retrieve/pii/S1094202517300546.

Mariacristina De Nardi, Eric French, and John B. Jones. Why do the elderly save? the role of medical expenses. *Journal of political economy*, 118(1):39–75, 2010.

Mariacristina De Nardi, Eric French, John Bailey Jones, and Rory McGee. Why do couples and singles save during retirement? household heterogeneity and its aggregate implications. *Journal of Political Economy*, 133(3):750–792, 2025.

Mariacristina DeNardi and Giulio Fella. Saving and wealth inequality. 2017.

Samuel L. Dickman, Steffie Woolhandler, Jacob Bor, Danny McCormick, David H. Bor, and David U. Himmelstein. Health spending for low-, middle-, and high-income americans, 1963–2012. *Health Affairs*, 35(7):1189–1196, 2016.

Samuel L. Dickman, David U. Himmelstein, and Steffie Woolhandler. Inequality and the health-care system in the usa. *Lancet*, 389(10077):1431–1441, 2017.

Andreas Fagereng, Martin Blomhoff Holm, Benjamin Moll, and Gisle Natvik. Saving Behavior Across the Wealth Distribution: The Importance of Capital Gains. Technical Report w26588, National Bureau of Economic Research, Cambridge, MA, December 2019. URL http://www.nber.org/papers/w26588.pdf.

Eric French and Elaine Kelly. Medical spending around the developed world. *Fiscal Studies*, 37 (3–4):327–344, 2016.

Xavier Gabaix. Power laws in economics and finance. *Annu. Rev. Econ.*, 1(1):255–294, 2009.

Fatih Guvenen, Fatih Karahan, Serdar Ozkan, and Jae Song. What do data on millions of us workers reveal about life-cycle earnings risk? Technical report, National Bureau of Economic Research, 2015.

Robert E. Hall and Charles I. Jones. The value of life and the rise in health spending. *Quarterly Journal of Economics*, 122(1), 2007.

Health and Retirement Study. Health and Retirement Study, public use dataset. Produced and distributed by the University of Michigan with funding from the National Institute on Aging (grant U01AG009740), Ann Arbor, MI, 2022. URL https://hrs.isr.umich.edu/.

Roozbeh Hosseini. Adverse selection in the annuity market and the role for social security. *Journal of Political Economy*, 123(4):941–984, 2015.

R Glenn Hubbard, Jonathan Skinner, and Stephen P Zeldes. Precautionary saving and social insurance. *Journal of political Economy*, 103(2):360–399, 1995.

Mark Huggett. The risk-free rate in heterogeneous-agent incomplete-insurance economies. *Journal of Economic Dynamics and Control*, 17:953–969, 1993.

Mark Huggett. Wealth distribution in life-cycle economies. *Journal of Monetary Economics*, 38(3): 469–494, 1996.

F Thomas Juster, James P Smith, and Frank Stafford. The measurement and structure of household wealth. *Labour Economics*, 6(2):253–275, 1999.

Karen A Kopecky and Tatyana Koreshkova. The impact of medical and nursing home expenses on savings. *American Economic Journal: Macroeconomics*, 6(3):29–72, 2014.

Lars Ljungqvist and Thomas J. Sargent. *Recursive Macroeconomic Theory*. MIT Press, Cambridge, 2012.

OECD. *Society at a Glance 2024: OECD Social Indicators*. OECD Publishing, Paris, 2024. doi: 10.1787/918d8db3-en. URL https://doi.org/10.1787/918d8db3-en.

Serdar Ozkan. Preventive vs curative medicine: a macroeconomic analysis of health care over the life cycle. 2014.

RAND Center for the Study of Aging. RAND HRS Longitudinal File 2018 (V2). Produced by the RAND Center for the Study of Aging, with funding from the National Institute on Aging and the Social Security Administration, Santa Monica, CA, 2022.

Emmanuel Saez and Stefanie Stantcheva. Generalized social marginal welfare weights for optimal tax theory. 106(1):24–45, 2016.

Emmanuel Saez and Gabriel Zucman. Wealth inequality in the united states since 1913: Evidence from capitalized income tax data. *The Quarterly Journal of Economics*, 131(2):519–578, 2016.

Jack Schechtman. An income fluctuation problem. *Journal of Economic Theory*, 12(2):218–241, 1976.

Jack Schechtman and Vera Escudero. Some results on "an income fluctuation problem". *Journal of Economic Theory*, 16(2):151–166, 1977.

Statistics Canada. Distributions of household economic accounts for income, consumption, saving and wealth of canadian households, first quarter 2025. https://www150.statcan.gc.ca/n1/daily-quotidien/250716/dq250716a-eng.pdf, July 16 2025. Accessed: October 5, 2025.

Jeffrey M Wooldridge. *Econometric analysis of cross section and panel data*. MIT press, 2010.